YOUNG SHIRE COUNCIL

ENGINEERING GUIDELINES FOR SUBDIVISIONS & DEVELOPMENTS

PART 3

Guidelines for Stormwater Drainage Design

Prepared by Engineering & Technical
Services Department
Young Shire Council

February, 1994

PART 3

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1. INTRODUCTION

This document outlines Young Shire Council's recommended practice for drainage design. All drainage design for Council should be carried out in accordance with this document. It is in no way a comprehensive "Design Manual" and is intended to be read in conjunction with and as a supplement to the 1987 edition of "Australian Rainfall and Run-Off" (referred to as "A R & R). The Design Co-Efficients defined in the Tables and Figures within this Manual are applicable only to the town of Young.

The "Major/Minor" concept is adopted for urban drainage design. The "Minor" system refers to the underground piped system, designed to an Average Recurrence Interval as determined in Section 3. The "Major" system refers to overland flow paths which are to be designed to convey major storm flows when the capacity of the minor system is exceeded.

Although the parameters and techniques for Flow Estimation in urban catchments, as set out in this Manual, generally refer to the Rational Method, it may, depending on catchment characteristics, be more appropriate to use a Unit Hydrograph or Non-Linear Run-Off Routing Model. The advantages and disadvantages of Unit Hydrographs has been more fully explained on pp. 255 and 256 of A R & R 1987.

All references to the Director should be interpreted as referring to the Engineering and Technical Services Director or his nominated representative.

Part 3 of the Guidelines for Subdivisions and Developments outlines Council's Guidelines for design of Stormwater Drainage Systems.

Part 1 - General Requirements
Part 2 - Guidelines for Design of Roads
Part 4 - Guidelines for Design of Water Reticulation
Part 5 - Guidelines for Design of Sewerage Reticulation
Part 6 - Guidelines for Landscaping, Erosion and Dust Control
Part 7 - Guidelines for Testing

2. DESIGN PROCEDURE

Listed below is the suggested procedure for urban stormwater drainage design. The steps refer to the general order in which a drainage design proceeds.

1. Determine Design Average Recurrence Intervals

2. Preliminary Layout of Proposed Drainage

3. Calculation of Minor System Flow Rates:-

    3.1 Design Rainfall Intensities
    3.2 Times of Concentration
    3.3 Run-off Co-efficients
    3.4 Sub-area Discharge
    3.5 Partial Area Considerations

4. Design of Pit Inlets

5. Assessment of Major System Flows

6. Hydraulic Design of Minor System

3. DESIGN RECURRENCE INTERVALS

3.1 Minor System Drainage

Road Drainage

Land Use	A.R.I.
Rural	15 years
Rural Residential	5 years
Urban Residential	5 years
Commercial	10 years
Industrial	10 years

3.2 Major System Drainage

A check is to be undertaken to ensure that the 100 year A.R.I. flowrate has a safe "escape route" when the minor system fails. Major system drainage is not to be confused with trunk drainage.

3.3 Trunk Drainage

All trunk drainage systems are to be designed to a 100 year A.R.I. flowrate for the Critical Storm, with a 0.5 metre freeboard on top of calculated top water levels. Any major retention structures should be checked for Half Probable Maximum Flood flows so that damage, resulting from failure of the structure, is minimised.

Modification of these values will depend on individual cases, eg. Freeboard may be increased or reduced where waterway area is such that the sensitivity of the floodway to changes in the flow parameters, is indicative of a corresponding large or small change in flow depth and velocity; additionally, it may be necessary for a structure to be checked for a Full Probable Maximum Flood in areas where failure could significantly increase the danger to life and property.

4. PRELIMINARY LAYOUT OF PROPOSED DRAINAGE

An assessment of the topography will determine the location of proposed drainage paths. Once the location of a proposed network is defined, trial pit locations should be specified. Generally, pits should be spaced so that there is minimal bypass flows. The width of flow in the gutter should not exceed 2.5 metres and pits should be spaced no further apart than 75 metres. Pits at road intersections should be located so as to limit flow around the kerb return to a maximum of 20 litres/second. An approximate procedure for locating pits is detailed in "Technical Note 2" in A.R. & R. (page 300). Also refer to Section 6 regarding pit inlet design and location.

Catchment areas to each pit can then be determined from contour information and proposed property boundaries. A site inspection should always be made to check the contour information and assess the likelihood of any flow path deviations which may occur as a consequence of existing or proposed developments. Changes to flow paths can occur as a result of the construction of fences, retaining walls, buildings etc. after the construction phase of the subdivision. The impact of these changes should be considered at the design stage.

Sub-areas discharges can be calculated using the procedures detailed in Section 5. Major system flow paths should be defined at this stage, and analysed according to the procedures detailed in Section 7.

5. CALCULATION OF MINOR STORMWATER FLOWRATES

5.1 Design Rainfall Intensities

Rainfall intensities adopted for design are those issued by the Bureau of Meteorology for Young, and these should apply for the full Young Shire Council Area. A copy of the I.F.D. Table follows. (See insert No 5.1).

5.1.1 Rainfall Intensity Diagram

(See following insert No 5.2).

5.2 Times of Concentration

Times of concentration for each sub-catchment should be determined using the Kinematic Wave Equation, as detailed in Technical Note 3 (Page 300) of AR&R.

$$t = 6.94\,(Ln^{*})^{0.6}/I^{0.4}S^{0.3} \quad \text{..... } \textit{Kinematic Wave Equation}$$

The Kinematic Wave Equation is very sensitive to slope and the Retardance Coefficient "n*", these should be estimated carefully. Recommended Retardance Coefficients are listed in Table 5.3.

TABLE 5.3

Surface Retardance Coefficients (n*)

LAND USE	RETARDANCE COEFFICIENTS
Road/Paved Areas Only	0.01
Normal Residential	0.08
Medium Density Residential	0.06
Industrial/Commercial	0.04
Parkland	0.15
Open Space (Natural Bushland)	0.3

Note: Minimum Tc = 6 minutes
Maximum Tc = 20 minutes
Where Tc is in excess of 14 minutes, it should be necessary to validate the use of such a Time in the calculations.

LOCATION 34.325 S 148.325 E * NEAR.. YOUNG ISSUED 30TH APRIL 1991 REF.-FN3476

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* ENSURE THE COORDINATES ARE THOSE REQUIRED, SINCE DATA IS BASED ON THESE AND NOT THE LOCATION NAME

LIST OF COEFFICIENTS TO EQUATIONS OF THE FORM

$$\ln(I) = a + b*(\ln(T)) + c*(\ln(T))**2 + d*(\ln(T))**3 + e*(\ln(T))**4 + f*(\ln(T))**5 + g*(\ln(T))**6$$

T = TIME IN HOURS I = INTENSITY IN MILLIMETRES PER HOUR

TABLE 5.1

RETURN PERIOD (YEARS)	a	b	c	d	e	f	g
1	2.8222	-0.6317	-0.0555	0.00787	0.002780	-0.0002141	-0.0000766
2	3.0876	-0.6389	-0.0558	0.00836	0.002774	-0.0002535	-0.0000726
5	3.3567	-0.6540	-0.0573	0.00843	0.002849	-0.0002585	-0.0000746
10	3.4961	-0.6618	-0.0578	0.00800	0.002915	-0.0001910	-0.0000891
20	3.6555	-0.6704	-0.0581	0.00906	0.002746	-0.0003121	-0.0000631
50	3.8413	-0.6784	-0.0587	0.00880	0.002832	-0.0002686	-0.0000748
100	3.9686	-0.6841	-0.0591	0.00895	0.002828	-0.0002872	-0.0000719

TABLE 5.2. RAINFALL INTENSITY IN MM/HR FOR VARIOUS DURATIONS AND RETURN PERIODS

DURATION (HOURS)	RETURN PERIOD 1 YEAR	2 YEARS	5 YEARS	10 YEARS	20 YEARS	50 YEARS	100 YEARS
0.083	56.6	74.8	101.	118.	140.	172.	197.
0.100	52.8	69.7	93.8	110.	130.	159.	183.
0.167	43.0	56.6	75.8	88.3	105.	128.	146.
0.333	31.3	41.1	54.6	63.3	74.8	90.8	104.
0.500	25.3	33.2	43.8	50.7	59.7	72.3	82.4
1.000	16.8	21.9	28.7	33.0	38.7	46.6	52.9
2.000	10.6	13.8	17.8	20.3	23.7	28.4	32.1
3.000	7.97	10.3	13.2	15.1	17.5	20.9	23.6
6.000	4.85	6.25	7.94	8.97	10.4	12.3	13.8
12.000	3.00	3.85	4.82	5.42	6.24	7.34	8.20
24.000	1.90	2.43	3.00	3.35	3.83	4.48	4.98
48.000	1.20	1.52	1.86	2.06	2.34	2.71	3.00
72.000	.884	1.12	1.35	1.48	1.68	1.93	2.13

INSERT No 5.2.

DESIGN RAINFALL INTENSITY DIAGRAM

LOCATION 34.325 S 148.325 E * NEAR.. YOUNG

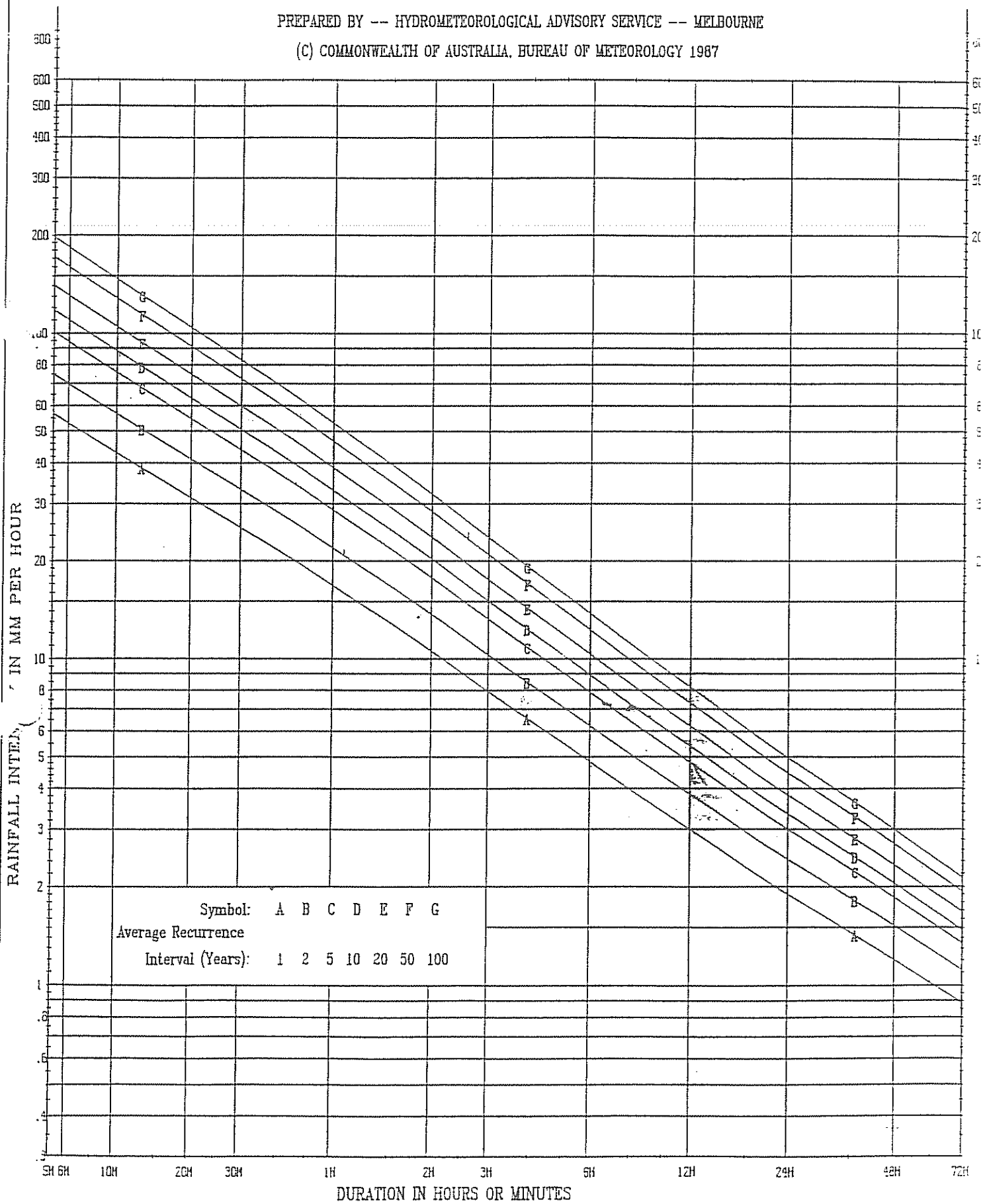
* ENSURE THE COORDINATES ARE THOSE REQUIRED.
SINCE DATA IS BASED ON THESE AND NOT THE LOCATION NAME.

ISSUED 30TH APRIL 1991 REF.-FN3476

IRBN DATA 22.55, 3.89, 1.12, 44.93, 7.01, 1.85, 0.249, 1HF1

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5.3 Run-Off Co-Efficients

Run-off coefficients "C" are to be determined in accordance with Section 14.5.5 (iii) (Page 307) of AR&R. The following equations apply for Young :-

$$Cy = Fy \times C10$$

where: $C_{10} = 0.9 \times f + 0.2064 \times (1 - f)$

These parameters are defined below:-

Cy: Run-off Coefficient for recurrence interval "y" (years)
C10: 10 year ARI Run-off Coefficient
Fy: Frequency Factor - see Table 5.4
f: Fraction Impervious - see Table 5.5

Run-off coefficients should be estimated separately for each land use. Table 5.6 is a quick reference table giving run-off coefficients in Young for the land uses listed in Table 5.5

Where an analysis is to proceed which requires the estimation of Initial and Continuing Loss Rates, these should be set to ZERO (0). That is, there should be no Initial and no Continuing Loss incorporated in any analysis.

Past experience suggests that major flooding in Young occurs when the catchment is already saturated and the Loss Rates are negligible.

TABLE 5.4

Frequency Factors for Run-off Coefficients

Recurrence Interval	Fy
1	0.8
2	0.85
5	0.95
10	1.0
20	1.05
50	1.15
100	1.2

TABLE 5.5

Fraction Impervious for Run-off Coefficients

Land Use	f
Normal Residential Lot Only	0.6
Normal Residential Lot Including Half Road	0.65
Half Width Road Reserve	0.8
Medium Density Residential Lot (villas etc)	0.7
Commercial Areas	0.95
Industrial Areas	0.9
Public Recreation Areas	0.4
Open Space (Natural Bushland)	0.35

TABLE 5.6

Quick Reference Table
Run-Off Co-Efficients for Young

Land Use	Recurrence Interval			
	5	10	20	100
Residential Lot Only	0.59	0.62	0.65	0.74
Residential incl. Half Road	0.63	0.66	0.69	0.79
Half Width Road Reserve	0.72	0.76	0.80	0.91
Medium Density Residential Lot	0.66	0.69	0.72	0.83
Commercial Areas	0.83	0.87	0.91	1.04
Industrial Areas	0.79	0.83	0.87	1.00
Public Recreation Areas	0.46	0.48	0.50	0.58
Open Space (Natural Bushland)	0.43	0.45	0.47	0.54

5.4 Sub-Area Discharge

Discharge for each sub-catchment may be calculated using the Rational Method formula:

$$Q = C \times I \times A / 0.36 \qquad (l/s)$$

"Hydrological Design Sheet 1" (copy attached) sets out the preferred format for these calculations. Technical Note 6 (page 312) of AR&R details a worked example for these calculations. The following points should be noted:

* The rainfall intensity adopted (column 8) is for the total flow time.

* A "C" value (column 9) and sub area (column 10) is calculated for each land use type.

* The discharge for the sub-catchment (column 13) is calculated by summing the CA's column 12) and multiplying by (I / 0.36).

"Hydrological Design Sheet 2" (copy attached) can then be used to calculate the flows along each reach of the pipe system. Where calculations are proposed to be presented as a computer hard copy, it should be necessary to obtain approval from the Director prior to doing so. Hand calculations should be submitted in the format shown on the attached Sheets.

5.5 Partial Area Flows

In urban catchments, it is probable that a greater flow rate may be obtained by applying the Rational Method to a lower part of the catchment with a time of concentration less than the full area travel time. These partial areas effects commonly occur when large paved areas are directly connected to the pipe inlet, and the sub-catchment discharge is based on a larger pervious area. Similarly, partial area effects can also occur, where a large open space catchment contributes to an urban catchment, with a Time of Concentration substantially different to the urban catchment.

In areas where this may be critical, such as industrial or high density residential development, a partial area check, based on times of concentration of impervious areas directly connected to the pipe system, is necessary.

6. PIT INLET DESIGN

6.1 Pit Location

The following criteria governs the location of pits in roadways, for the design average recurrence interval:

(i) Inlet Pits are to be spaced so that flow width should not exceed 2.5m (may not be applicable in the upper reaches of the catchment) for the 5 year ARI Storm.

(ii) Gutter flows should at no point overtop the kerb.

(iii) Velocity x depth should not exceed 0.4 (5 year ARI)

(iv) Bypass from any pit on grade is not to exceed 15% of the total flow at the pit (Full Capture Desirable).

(v) Maximum spacing between pits should not exceed 75m.

(vi) Where flows in the gutter are >= 20 l/s and/or >= 1m in width adjacent to the upstream Kerb Return Tangent Point, it will be necessary to intercept these flows with a Kerb Inlet Pit.

(vii) The location of gully pits on curves is to be avoided and they are not to be placed in line with the normal passage of pedestrians.

Technical Note 4 (page 304) of AR&R details a method for calculating gutter and road flows.

6.2 Inlet Design

Once the sub-catchment flows are known, pit inlets should be designed in accordance with the charts contained on the following pages. All new pit inlets are to be constructed using welded steel ("Weldlok" or similar) type grates which allow access to the pit. On grade, percentage capture by grates is mainly dependant on lintel size.

Hand calculations for the design of pit inlets is to be tabulated on "Hydrological Design Sheet 1" (column 16 to 20).

Sag pits are to be designed based on a depth of ponding, up to the top of the kerb. Ponding depths can be calculated using the design charts given in Section 19, or the formulae on page 303 of AR&R with the adopted blocking factors. Minimum internal lintel size in a sag should be 2.4m.

7. DRAINAGE PIT DESIGN

(a) General

(i) Standard pits should be provided in drainage lines at all changes in grade, level or direction and at all pipe junctions;

(ii) The minimum clearance from the top of the manhole to the design water level in the pit should be 150mm;

(iii) Pipe junctions where the deflection angle of the major flow is 90° should be avoided;

(iv) Pipe grading across pits should be designed on the following basis:-

(a) No change in direction or diameter - minimum 50mm;
(b) No change in diameter but direction change - minimum 70mm;
(c) Changes in diameter should be graded obvert to obvert;

(v) Where the depth of the pit exceeds 1.2 metres, standard galvanised or other approved step-irons are to be provided at a spacing of 300mm to provide access for inspection and cleaning;

(vi) Every endeavour is to be made to maintain flow velocities through pits and excessive drops will not be permitted;

(vii) Pits are to be located and constructed in accordance with Council's drawings. Precast pits may be used if the prior approval to the type and design is obtained from the Director;

(viii) Pipe diameter 750mm or greater will require radius bends without pits where substantial change of direction is required.

(b) Inlet Pits

Inlet pits are to be constructed in accordance with Council's Drawing No's 6480 to 6484 inclusive.

(c) Angle Pits and Junction Pits

Angle and junction pits are to be constructed in accordance with Council's Drawing No's 6485 and 6486 respectively.

(d) Footpath and Field Surface Inlet Pits

Where footpath or field surface inlet pits are required, they are to be constructed in accordance with Council's Drawing No's 6487 and 6488 respectively.

(e) Surcharge Surface Inlet Pit

At the upstream end of drainage pipelines in public reserves, gullies, etc. and in subdivisions involving staged construction an inlet pit with Weldlok or similar surcharge surface inlet pit cover is to be provided in accordance with Council's Drawing No. 6489.

8. ASSESSMENT OF MAJOR SYSTEM FLOWS

All urban drainage designs should incorporate an assessment of major system flows. The aim of this check is to ensure that a safe and adequate "escape route" is achieved for large storm events.

The Rational Method may be used to estimate major system flows for critical points in the drainage system. An ARI of 100 years should be used for this and the difference between the minor system flow and the 100 year ARI flow should be the basis upon which the major system flow path should be designed.

The roads and pathways will generally form the flow path by which the major system flows are routed, either to the trunk drainage system or to a lowpoint with sufficient hydraulic capacity to capture the flows. Special consideration needs to be given to trapped lowpoints where the overland flowpath may divert surcharge into properties. This is especially important when designing "Downhill" cul-de-sac and kerb returns adjacent to a sag vertical curve. In the former case the overland flowpath could possibly incorporate a depressed pathway with reverse crossfall in the footpath. In the latter case consideration should be given to grading the kerb return such that water flows around the return and away before it breaks over the top of kerb at the lowpoint. Technical Note 4 (page 304) of AR&R details a method for calculating gutter and road flows.

If the roadway capacity is inadequate, an estimate should be made of the capacity of the pipe system under major flow conditions. Pit capacities should be calculated using the appropriate blocking factors, and pipe capacities estimated with trial diameters and head levels no greater than 150m below the surface levels/invert of kerb (applicable up to the design ARI for the respective pipeline reach). The computer program listed on page 318 of AR&R will calculate pipe capacities for this purpose.

The major flows must have an overland flow path such that all habitable floor levels are a minimum of 0.3m above the 100 year ARI Top Water Level. The freeboard requirement may be varied by the Director on consideration of the sensitivity of the floodway parameters to the flows.

Technical Note 6 (page 316) in AR&R details a procedure for checking major system flows. A copy of "Hydrological Design Sheet 3" the preferred format for checking major systems, is on the attached.

9. PIPELINE DESIGN

9.1 General Requirements

(a) Pipelines in roadways should have a minimum diameter of 300mm in urban areas and 450mm in Rural areas.

(b) In general, pipelines should cross roads at right angles to the road centreline.

(c) Drainage should not be carried a greater distance than 75 metres in pipelines without adequate inspection manholes. Where the pipeline diameter exceeds 1200mm, this distance may be increased to 100 metres.

(d) For single cell pipe systems, a downstream pipe of smaller diameter than the upstream pipe will not be permitted for pipes with diameter less than 900mm.

(e) A desirable minimum grade of 1% of all pipelines is preferred for self-cleansing under low flow velocities. Flow velocities should be limited to a minimum of 0.6m/s for self-cleansing purposes.

(f) The inlet and outlet drains to pipelines should be carefully designed so as to avoid either scouring or silting velocities during storm flows, and adequate scour protection satisfactory to the Director is to be provided at the outlet of all stormwater lines.

(g) Curved pipelines will not be permitted where the diameter of the pipeline is less than 750mm. Where curved pipelines are permitted, they are to be constructed using rubber ring jointed reinforced concrete pipes and installed strictly in accordance with the Manufacturer's recommended radii.

9.2 Headwalls

Concrete headwalls should be constructed for all culverts in accordance with Council's Drawing No. 7310.

9.3 Concrete Bulkheads

Concrete bulkheads should be constructed for all drainage lines exceeding 6% grade, at intervals not exceeding 15 metres, in accordance with Council's Drawing No. 7312.

9.4 Sub-Soil Drainage

Sub-soil drains may be required in certain locations, particularly as intercepting drains behind kerb and for drainage of the flexible pavement. Sub-soil drainage lines should be graded to suitable outlets such as stormwater pits.

10. HYDRAULIC DESIGN

The Hydraulic Grade Line method should be used for pipeline design. It is not the purpose of this document to give a detailed explanation of this method.

Technical Note 9 (page 330) of AR&R details the recommended procedures for Hydraulic Grade Line calculations (HGL). A copy of the "Hydraulic Design Sheet" is attached and this should be used for manual calculations submitted to Council. Computer analyses will be accepted where prior approval from the Director has been received. The initial pipe sizing should be performed on a reach by reach basis and final HGL checking of the system should then be performed from the system outlet working upstream. The Water Levels provided by the HGL calculations should be plotted on the pipeline Longitudinal Sections.

10.1 Recommendation Pipe Friction Coefficients

PIPE MATERIAL	MANNINGS "n"	COLEBROOK-WITE "k"
Concrete	0.012	0.60
UPVC	0.009	0.03

10.2 Tailwater Levels

The tailwater to be adopted will depend on the outflow conditions. Where determination of a tailwater level is in doubt, it should be necessary to confirm the value with the Director prior to proceeding:-

* for free outfalls, adopt the pipe obvert.

* for discharge into receiving waters, adopt a tailwater equivalent to the design ARI flood level.

* for discharge into existing systems where the hydraulic grade levels are unknown, adopt a tailwater 150mm below the natural surface/invert of kerb.

* for discharge into a point designed to surcharge, adopt a tailwater level equivalent to the height of surcharge.

10.3 Inlet Efficiency

An allowance of 150mm is to be adopted below the lowest point of the pit inlet/kerb invert, to effectively allow such inlets to act efficiently. Where this cannot be achieved it will be necessary to obtain the concurrence of Council's Engineer before proceeding.

10.4 Pit Loss Factors

Pit loss factors (k) should be calculated using the Tables from ARRB SR 34, 1986 (Tables 6.5 and 6.6). Where the pit configuration is not covered by these tables, the Missouri Charts, Hare Charts or Department of Housing Charts should be used.

11. DRAINAGE EASEMENTS

(a) Where it is intended to create drainage easements for a new road provided in a subdivision, a notation should appear on the engineering drawings and subdivision plan creating the easement or easements pursuant to Section 88B of the Conveyancing Act, 1919 as amended.

(b) Where stormwater drainage conveyed in a drainage easement, piped or otherwise, such as referred in (a) above would discharge onto land other than an existing drainage easement or public place, it should be the responsibility of the developer to obtain a drainage easement through such land, sufficient in dimension to convey the drainage to an easement, natural watercourse or public place, and to transfer easement rights thereover to Council. The linen plan of subdivision will not be released until the above requirements have been complied with, and all fees and contributions have been paid.

(c) Where a drainage easement lies within a development which does not involve the opening of a new road, the developer should transfer to Council any drainage easement provided in the subdivision and execute a transfer and grant of easement in favour of Council pursuant to Section 88B of the Conveyancing Act, 1919 as amended.

(d) Drainage easements are generally to be piped within the subdivision, if not otherwise approved by the Director.

(e) In parkland, pipes sufficient to carry a 1 in 1 year frequency storm and the dry weather flow should be installed plus an open grassed channel to meet additional design requirements. Should the provision of such piping be deemed excessive, the Director should determine Council's requirements.

(f) Where stormwater is drained through land classified as non-urban, open grassed channels may be considered.

(g) Drainage Outside the Subdivision

Depending upon the location of the development within a catchment, the developer will be responsible for stormwater discharged from the proposed development onto other lands.

DRAFT

(h) Width of Easements

(i) Interlot Drainage - 1.5 metres;

(ii) Piped Drainage Formula Width = (1.5 x depth of trench) + diameter of pipe Minimum Easement Width is 3.0 metres;

(iii) Open Channels - Formula = Top width of 100 year design flow with freeboard + 1 metre;

12. INTERLOT DRAINAGE SYSTEMS

(a) The installation of these pipes is deemed necessary where roofwater and surface water cannot be discharged directly to the street gutter. Interlot drainage systems are intended to collect both roofwater and surface water.

(b) Each lot served by an interlot drainage line should be provided with a grated inlet structure to permit the inlet of surface water.

(c) The minimum size pipe is to be 225mm diameter. Construction is to include manholes at intervals of not more than 75 metres.

Consideration will be given to the installation of a 150mm diameter pipe, where only one (1) lot is to be served.

(d) Private drainage lines are to be located within an easement 1.5 metres wide, granted in favour of the properties served by that line and, should be located in the higher rather than the lower property.

(e) At the most suitable point in each lot, an inlet pit is to be brought to surface level with an exposed lid. From the wall of the pit a 150mm junction is to be left for the connection of roofwater. The connection by the plumber of roofwater to this point is to be supervised by Council's Building Inspector. Interlot drainage pits are to be constructed in accordance with Council's Drawing No. 6490.

(f) Tables for design of interlot drainage are given in the following page.

Design of Private Stormwater Drainage in Subdivisions

A guide for the minimum size of pipe to be provided for private urban stormwater drainage lines is given in Tables 12.1 and 12.2.

TABLE 12.1

NO. OF LOTS	GRADE (%)									
	1.0	1.5	2.0	2.5	3.0	4.0	5.0	6.0	7.0	8.0
1	225	225	225	225	150	150	150	150	150	150
2	225	225	225	225	225	225	225	225	225	225
3	225	225	225	225	225	225	225	225	225	225
4	225	225	225	225	225	225	225	225	225	225
5	300	225	225	225	225	225	225	225	225	225
6	300	225	225	225	225	225	225	225	225	225
7	300	300	225	225	225	225	225	225	225	225
8	300	300	225	225	225	225	225	225	225	225
9	300	300	300	225	225	225	225	225	225	225
10	375	300	300	300	225	225	225	225	225	225
12	375	300	300	300	300	300	225	225	225	225
14	375	375	300	300	300	300	300	300	225	225
16	375	375	375	300	300	300	300	300	300	300

Design of Private Industrial Stormwater Drainage in Subdivisions

The minimum size of pipe to be provided for private industrial stormwater drainage lines should be in accordance with the table below:

TABLE 12.2

NO. OF LOTS	GRADE (%)									
	1.0	1.5	2.0	2.5	3.0	4.0	5.0	6.0	7.0	8.0
1	225	225	225	225	225	225	225	225	225	225
2	300	300	225	225	225	225	225	225	225	225
3	300	300	300	300	300	300	225	225	225	225
4	375	375	300	300	300	300	300	300	300	225
5	375	375	375	375	375	300	300	300	300	300
6	375	375	375	375	375	300	300	300	300	300
7	450	375	375	375	375	375	300	300	300	300
8	450	450	375	375	375	375	375	300	300	300
9	450	450	450	375	375	375	375	375	375	300
10	525	450	450	450	375	375	375	375	375	375
12	525	525	450	450	450	450	375	375	375	375
14	525	525	525	450	450	450	450	375	375	375
16	600	525	525	525	450	450	450	450	450	375
18	600	525	525	525	525	450	450	450	450	450
20	600	600	525	525	525	450	450	450	450	450

13. TRUNK DRAINAGE SYSTEMS

An example of the suggested procedure for trunk drainage design is contained in "Technical Note 10" (page 343) of AR&R.

Definition: As a guide a trunk drain should be considered to be a drain that would require a 1200mm diameter pipe or greater to convey the 1 in 5 year storm flow.

13.1 Calculation of Flows

Flows through a trunk drainage network should be calculated using an appropriate runoff-routing computer model as detailed in Chapter 9 of AR&R. Details of the model to be used are to be submitted to Council, along with IBM compatible disk copies of the input and output data. Should the model used not be IBM compatible, hard copies of input and output are required.

Details of the calibration of the model should be submitted to Council. The model should be calibrated against a known discharge (or one calculated using an alternative method) for the existing catchment conditions, then the impact of the flow from the development catchment conditions analysed. NOTE: Once calibrated, the model is also to be analysed for the impact on the drainage network of flows generated, without the use of Initial or Continuing Loss Rates, from the developed catchment.

13.2 Floodways/Open Channels

Trunk Drainage systems should be designed as "soft" systems i.e. grass lined channels with low flow pipelines or concrete lined inverts. Design criteria are:-

(a) Hydraulic Design

Open channels should be designed using backwater calculations (see enclosed calculation form attached). A freeboard of 0.5m above the 100 year ARI top water level is to be incorporated into the channel design (unless otherwise approved). The product of Velocity x Depth, where practicable, should not exceed 1. The table on the following page sets out a procedure for tabulating manual calculations.

(b) Low Flow Pipelines/Lined Inverts

Low Flow Pipelines are to be a minimum size of 375mm diameter, sized using a flow of 3 litres/sec/hectare for residential land uses and 10 litres/sec/hectare for industrial land uses. Alternatively, lined Low Flow inverts should accommodate flow equivalent to those of the Low Flow pipelines.

(c) Flow Velocities

Maximum flow velocities in grass-lined channels should be 2 m/s for a minimum 100 year ARI flow (unless otherwise specified). Designs should be based on sub-critical flow with a Froude Number no greater than 0.8.

(d) Scour Protection

Adequate scour protection should be designed for all discharge points into and out of the channel, or at any point in the channel where there is a significant change in flow conditions. Adjacent piped systems should connect to the Low Flow piped system as a continuation of the side lines pipe diameter. The connection point should be a surcharge pit capable of surcharging the side lines total discharge. No surcharge points should occur in the channel batters higher than the 100 year ARI Top Water Level.

(e) Stabilising

Suitable grasses should be used and maintained in open channels.

Open channels should be stabilised either by turfing or with a geotextile material with a minimum life expectancy of two (2) years. The details and specification for fixing of the geotextile should be submitted for the Director's approval.

All open channels should be provided with a minimum two (2) metre wide, 150mm thick concrete invert, where required by the Director.

(f) Batter Slopes

Batter slopes of grassed waterways should be a minimum of 1 (vertical) : 5 (horizontal). Minimum crossfalls in channels should be 2% with a depressed channel invert.

(g) Road Crossings

Road Culverts are to be designed in accordance with culvert hydraulics theory. The channel upstream should incorporate an appropriate allowance for afflux through the culvert. Appropriate scour protection should be included at the outlet.

(h) Curvature

The centreline curves should have a radius not less than twice the 100 year flow top width, with a minimum of 30 metres.

(i) Recommended Manning's Roughness Co-Efficients "n" for Open Channels, are as follows:-

	Recommended Values
1. Concrete Lined	0.013
2. Grassed	0.035
3. Earth (Clear)	0.02

14. RETENTION BASINS

Where downstream stormwater pipelines are inadequate and/or where economically feasible, retention basins are required at the discretion of the Director.

(a) Hydraulic Analysis

Generally, retention basins are to be designed such that the fully developed flow does not exceed the undeveloped flow. Each basin will have its own requirements, dependent upon downstream conditions. Hydraulic analysis is to be carried out using a suitable reservoir routing model as detailed in Chapter 9 of AR&R.

(b) Water Levels and Freeboard

There is no restriction on the depth of retention basins, however, each basin must be considered in light of the surrounding terrain and development. A freeboard of 500mm on the 100 year top water level is to be incorporated into the basin embankment (unless otherwise specified).

(c) Spillways

Spillways should be designed to pass extreme flood events without catastrophic failure of the embankment. Special consideration should be given to erosion protection on the spillways and the techniques proposed should require the approval of the Director prior to their final design.

(d) Batters

Grassed batters should be no steeper than 1 in 6. The minimum slope of the basin floor should be 1%.

(e) Grassing

All exposed areas are to be grassed.

(f) Basin Safety in Extreme Events

The performance of the structure in a storm event greater than a 100 year ARI storm is to be checked. A Minimum Half Possible Maximum Flood is to be routed through the structure to assess the impacts of the possible failure of the basin. Assessment of the Full Possible Maximum Flood should be necessary where a structural failure of the basin could cause catastrophic damage in adjoining areas.

15. RURAL - RESIDENTIAL SUBDIVISIONS

Calculation of flowrates in Rural - Residential subdivisions should be in accordance with Chapter 5, "Estimation of Peak Flows for small to medium sized Rural Catchments" of "Australian Rainfall and Runoff" 1987. Pipe drainage systems are required under all road reserves which should be designed as culverts.

Design guidelines for open-channel floodways are indicated in Section 13.

16. CULVERT DESIGN

Road culverts should be designed in accordance with culvert hydraulics theory i.e. the culvert capacity is determined by the flow conditions, depending on whether inlet control or outlet control governs.

Recommended design procedures are contained in Section 3 of the Concrete Pipe Association of Australia's publication: "Hydraulics of Precast Concrete Conduits - Hydraulic Design Manual". A suggested tabulation of the design calculations is attached.

Alternatively, M. Boyd has written short computer programs which can do these calculations, based on Head-Discharge Equations for culverts. These programs are called "CULVERT.Q" and "CULVERT HW", and may be purchased from the University of Wollongong.

17. DESIGN PRESENTATION

Generally, drainage designs are to be presented in the form shown in Chapter 14 of "Australian Rainfall and Runoff" 1987.

The following data is to be provided in association with a contoured catchment area plan, the scale of which should not be less than 1:2000:-

(i) Catchment areas, paths of overland flow and proposed and existing pipe layout, plan with all sub-areas, drainage lines and manholes numbered;

(ii) Area of catchment sub-areas (the area drained by each headwall, pit or any other pick-up point);

(iii) Length, slope and time of overland flow;

(iv) Length, slope and time of gutter flow;

(v) Sub-area and critical times of concentration;

(vi) Co-efficients of run-off;

(vii) Rainfall intensities;

(viii) Design discharges;

(ix) Slope of hydraulic gradients, corresponding velocities and times of flow;

(x) Hydraulic grade levels at manholes;

(xi) Size, length, slope and obvert levels of proposed drainage lines;

(xii) The accurate position and levels of all services crossed by any stormwater pipeline;

(xiii) Longitudinal sections for all drainage lines;

(xiv) Cross sections where necessary;

(xv) The minimum habitable floor levels for all lots adjacent to major stormwater flow paths should be clearly indicated;

(xvi) Tailwater Levels;

(xvii) Site contours should be indicated on the stormwater plan;

(xviii) Open channels should be detailed with cross sections at minimum ten (10) metre intervals;

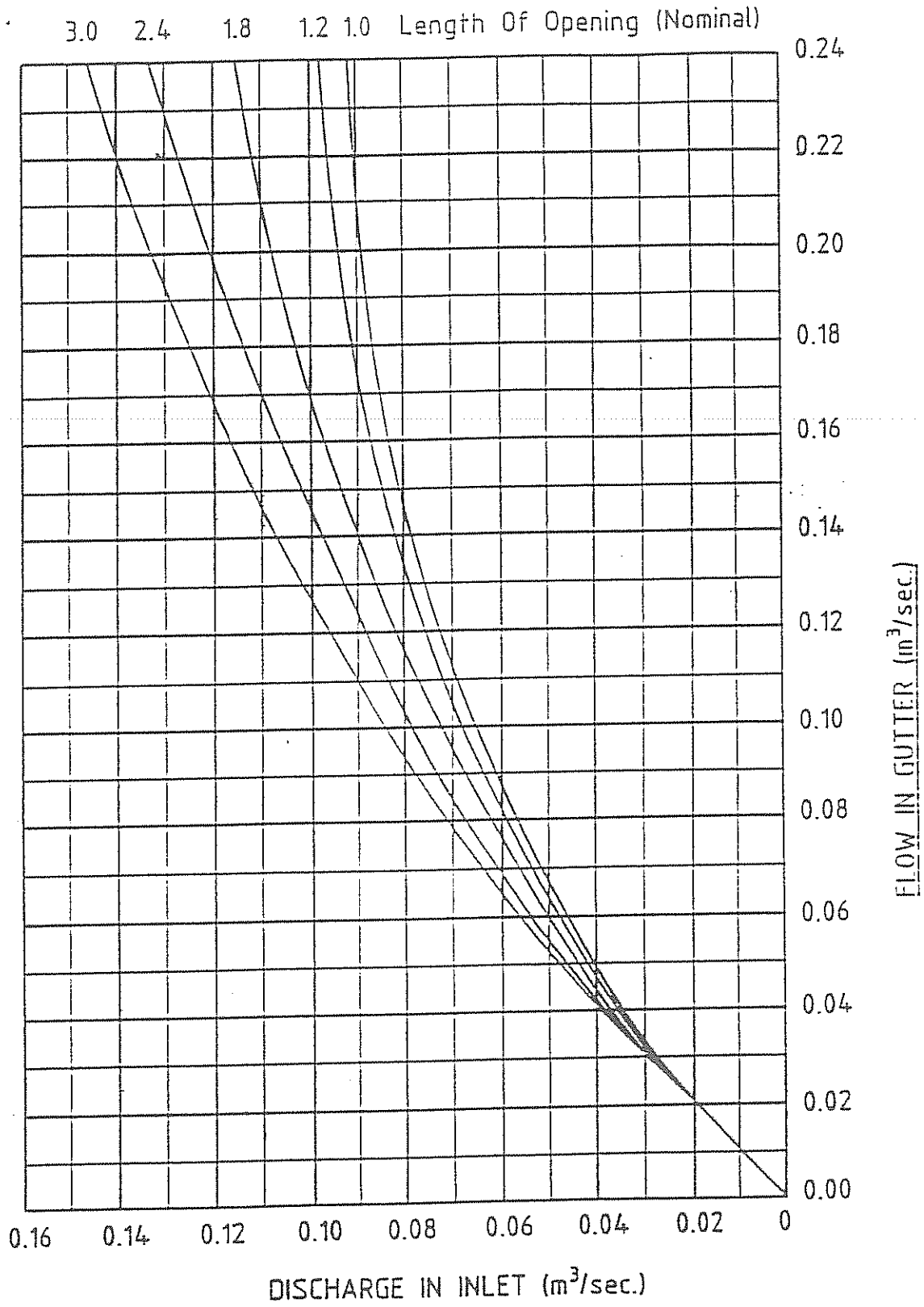
(xix) Longitudinal sections for all open channels with backwater profiles.

18. MATERIALS

Stormwater pipelines may be constructed from reinforced concrete pipes, fibre reinforced concrete pipes or uPVC pipes manufactured and installed in accordance with relevant Australian Standards.

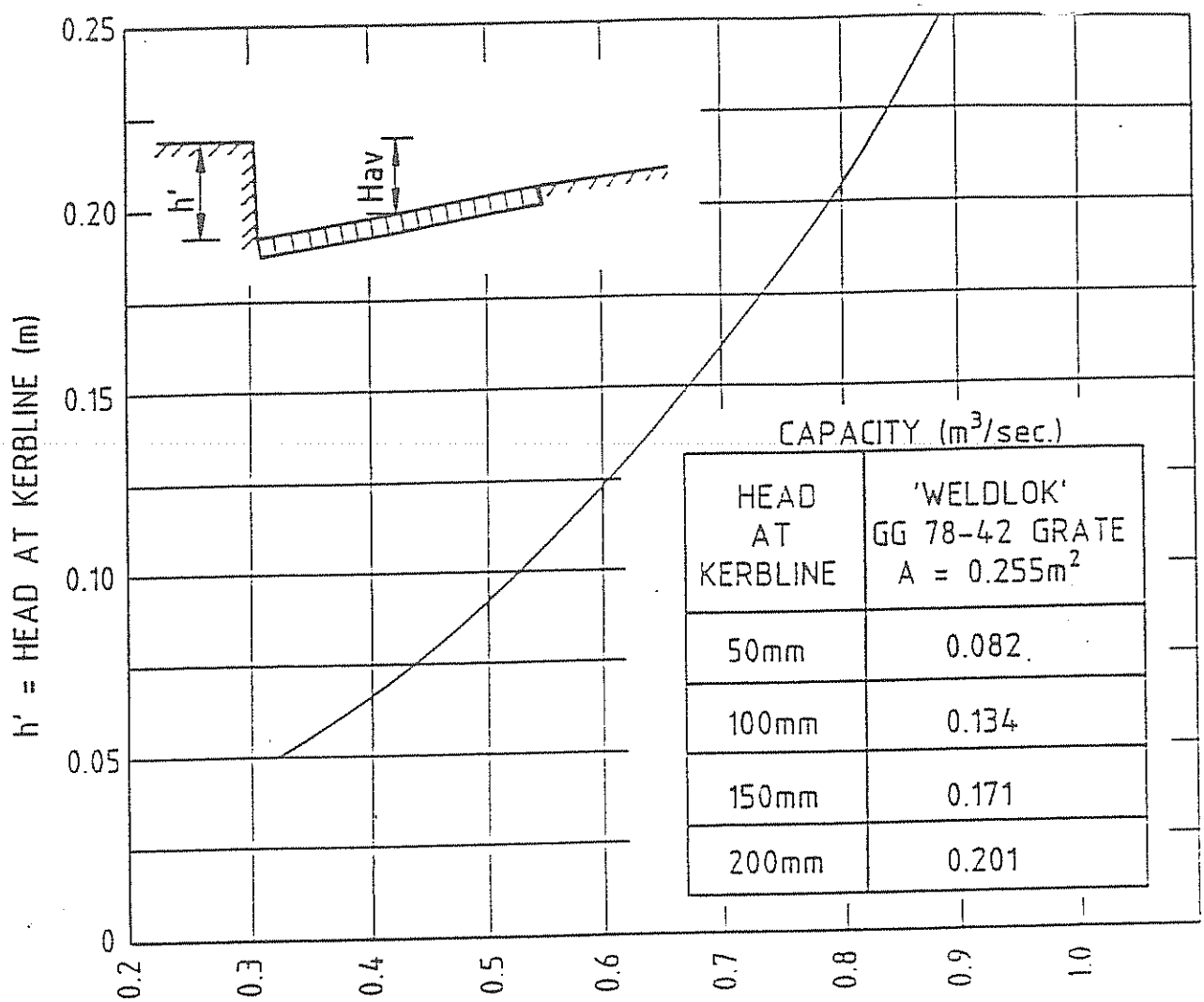
19. DRAWINGS AND GRAPHS

		Drawing No.
1.	Inlet Capacity Charts	
	(a) Kerb Inlet with Grate	6477
	(b) Grate in Sag	6478
	(c) Kerb Inlet in Sag	6479
2.	Stormwater, Kerb Inlet Pits with Combined Lintel/Grate	
	(a) Pipe located under Grate	6480
	(b) Pipe located under Kerb and Gutter	6481
	(c) Pipe located behind Kerb and Gutter	6482
	(d) Pipe located under Carriageway	6483
3.	Kerb Inlet Pit with Grate for Layback Kerb and Gutter	6484
4.	Angle Pit	6485
5.	Junction Pit	6486
6.	Footpath Surface Inlet Pit	6487
7.	Surface Inlet Pit	6488
8.	Surcharge Surface Inlet Pit	6489
9.	Interlot Stormwater Inlet Pits	6490
10.	Subsoil Drains	7305
11.	Pipe Bedding	7308
12.	Concrete Headwalls	7310
13.	Straight Concrete Headwalls and Bulkheads	7312



- Inlet Capacities Based On D.M.R. "Model Analysis To Determine Hydraulic Capacities Of Kerb Inlets And Gully Pit Gratings"- 1979
- Pit Capacities Interpolated Between D.M.R. Lintel Capacities Based On Clear Opening Of Lintel
- 3% Road Crossfall

YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
INLET CAPACITY CHART KERB INLET WITH GRATE (ON GRADE)	DRAWN TRACED CHECKED	DRAWING No. 6477



HEAD AT KERBLINE	'WELDLOK' GG 78-42 GRATE A = 0.255m²
50mm	0.082
100mm	0.134
150mm	0.171
200mm	0.201

Q: INLET CAPACITY IN m^3/sec. PER SQUARE METRE OF CLEAR OPENING (INCLUDING 30% REDUCTION FOR CLOGGING)

NOTE: $Q = 0.6\ A\ C\sqrt{2gh}\ \ m^3/sec.$

WHERE: A = Area Of Clear Opening In m^2

C = Clogging Factor = 0.7 (30% Reduction For Clogging)

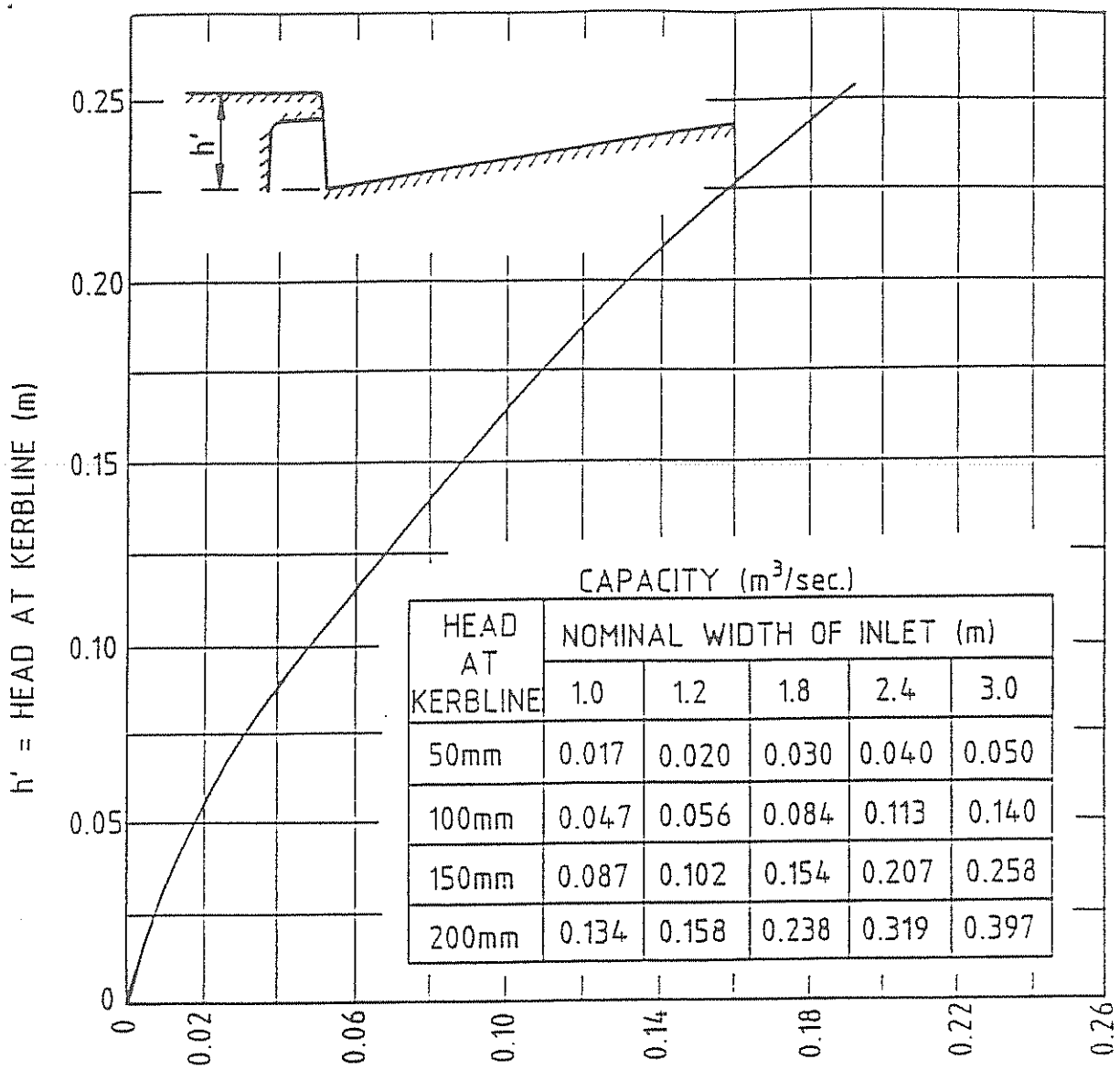
g = Gravitational Acceleration

h' = Head At Kerbline In Metres

Hav = h' - 0.020m

h = Hav

YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
INLET CAPACITY CHART GRATE IN SAG	DRAWN / TRACED / CHECKED	DRAWING No. 6478



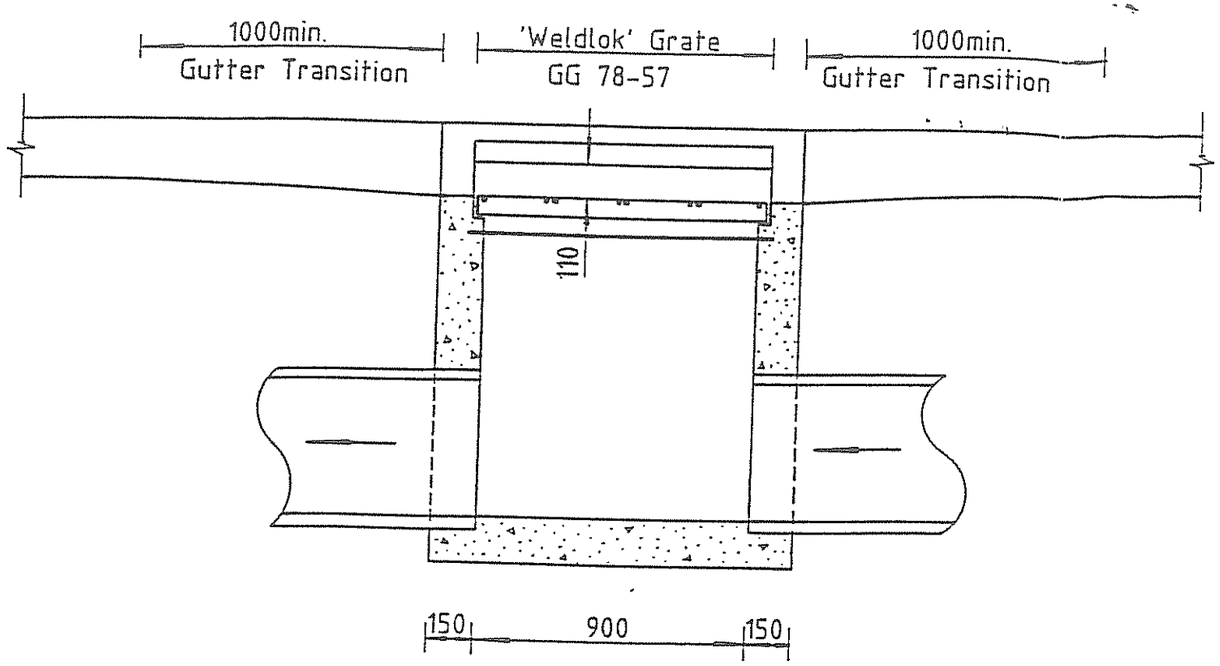
HEAD AT KERBLINE	NOMINAL WIDTH OF INLET (m)				
	1.0	1.2	1.8	2.4	3.0
50mm	0.017	0.020	0.030	0.040	0.050
100mm	0.047	0.056	0.084	0.113	0.140
150mm	0.087	0.102	0.154	0.207	0.258
200mm	0.134	0.158	0.238	0.319	0.397

Q: INLET CAPACITY IN m^3/sec. PER METRE LENGTH OF OPENING.
(INCLUDING 10% REDUCTION FOR CLOGGING)

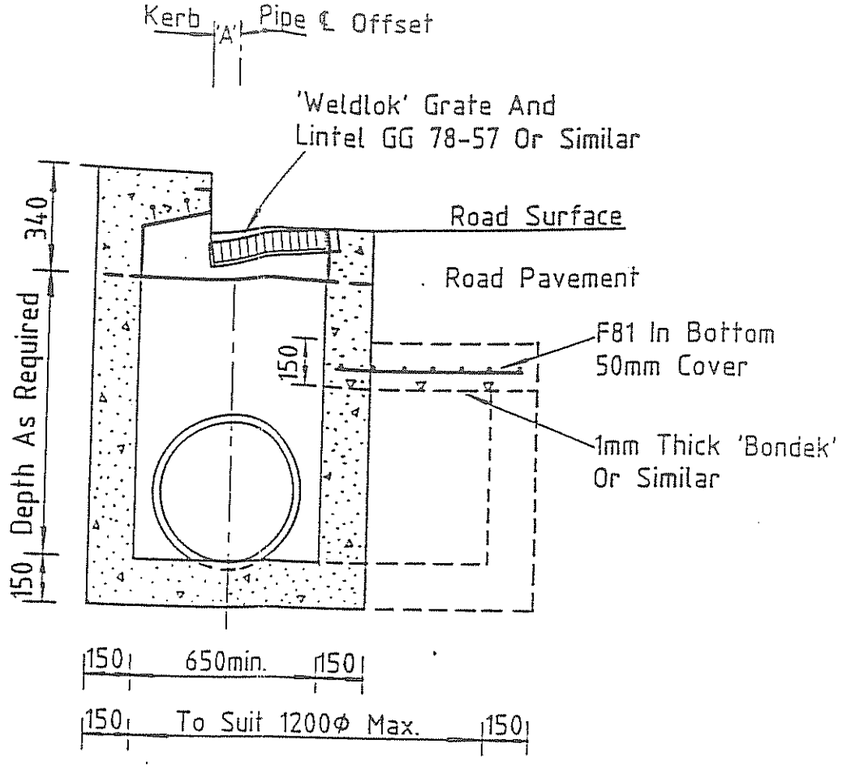
NOTE: $Q = 1.66\ Lh'^{3/2} \times C$

WHERE: L = Width Of Inlet In Metres
h' = Head At Kerbline In Metres
C = Clogging Factor = 0.9 (10% Reduction For Clogging)

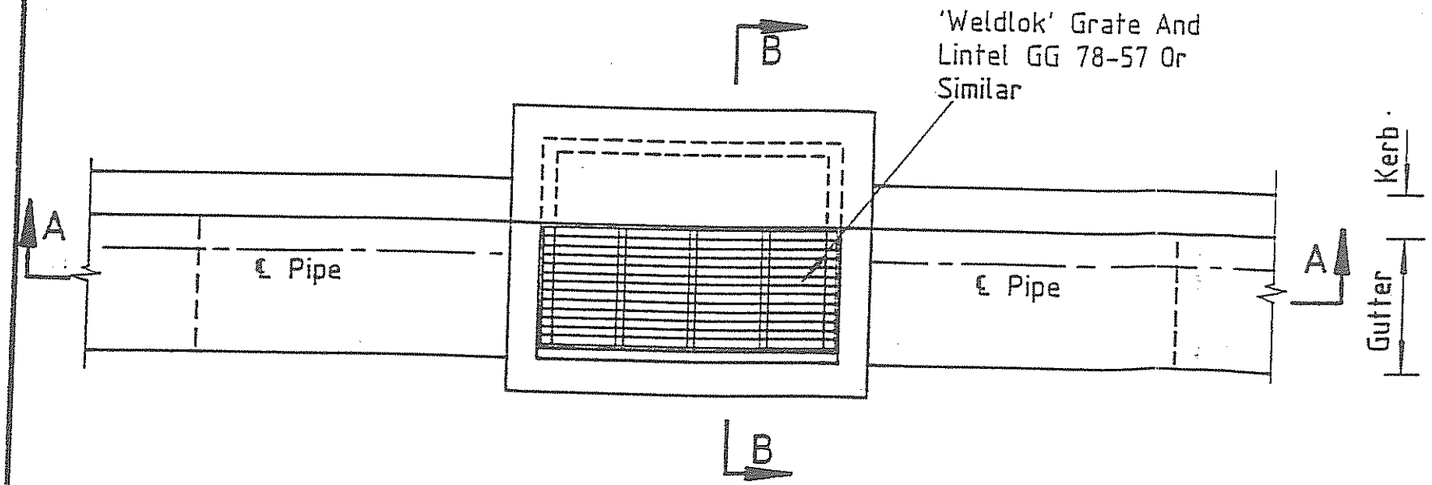
YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
INLET CAPACITY CHART KERB INLET IN SAG (NO GRATE)	DRAWN TRACED CHECKED	DRAWING No. 6479



SECTION A-A



SECTION B-B



PLAN

GENERAL NOTES

Concrete 20MPa

Step Irons To Be Provided @ 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

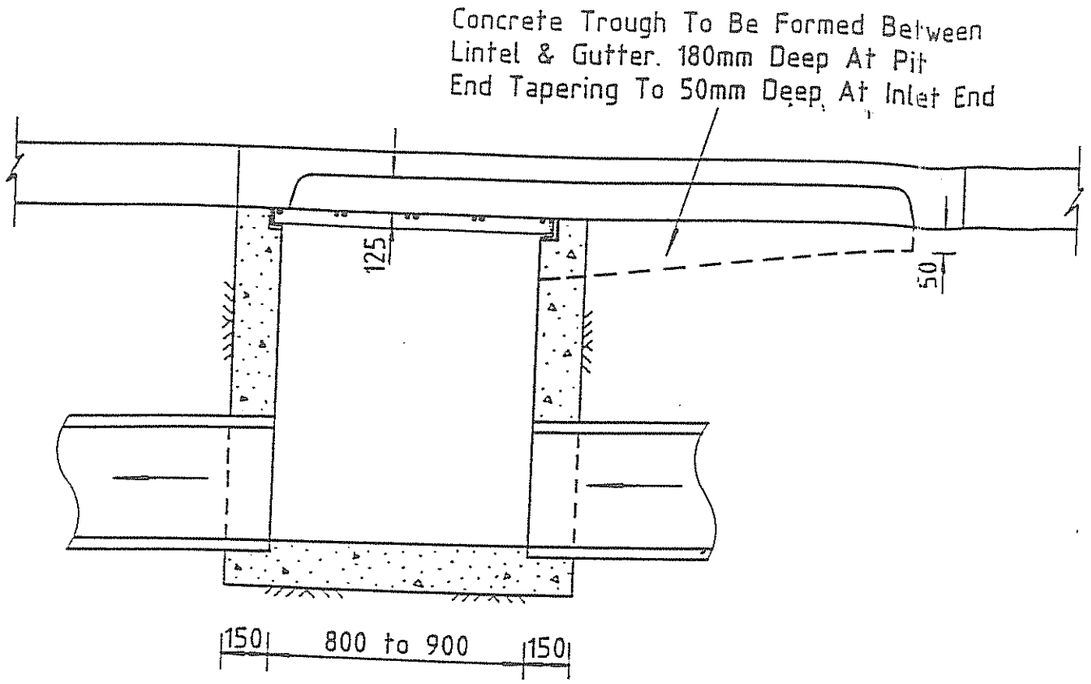
PIPE ℄ OFFSETS FROM KERB

PIPE ϕ	DISTANCE 'A'	PIPE ϕ	DISTANCE 'A'
375	100mm	750	200mm
450	100mm	825	250mm
525	100mm	900	300mm
600	150mm	1050	350mm
675	150mm	1200	450mm

YOUNG SHIRE COUNCIL

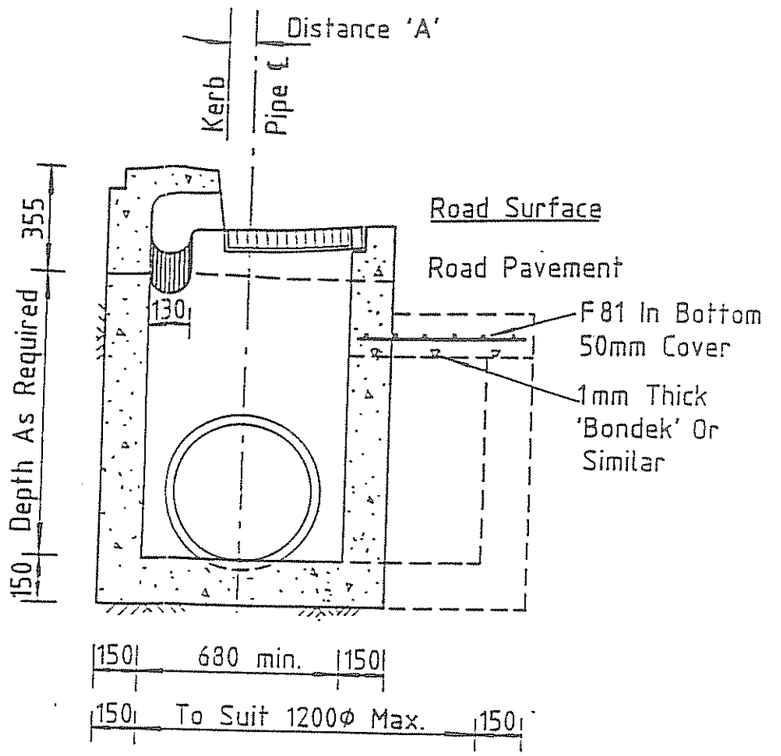
KERB INLET PIT WITH COMBINED LINTEL/GRATE

SCALES AS SHOWN	W & T SERVICES DIR.	DATE
DRAWN		
TRACED	A3	DRAWING No. 6480
CHECKED		

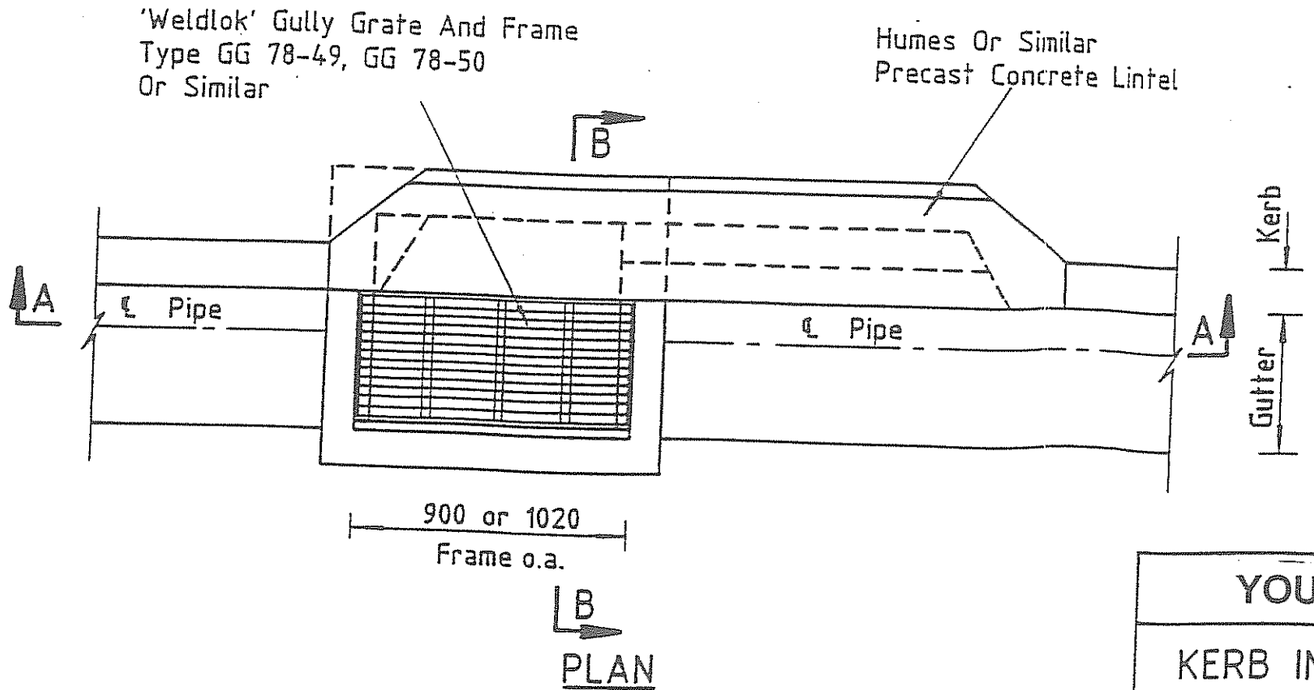


SECTION A-A

Note: Kerb Inlet To Be Adjusted As Required To Suit 'Weldlok' Grate & Lintel GG 78-57 Or Similar



SECTION B-B



PLAN

GENERAL NOTES

Concrete 20MPa

Step Irons To Be Provided @ 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

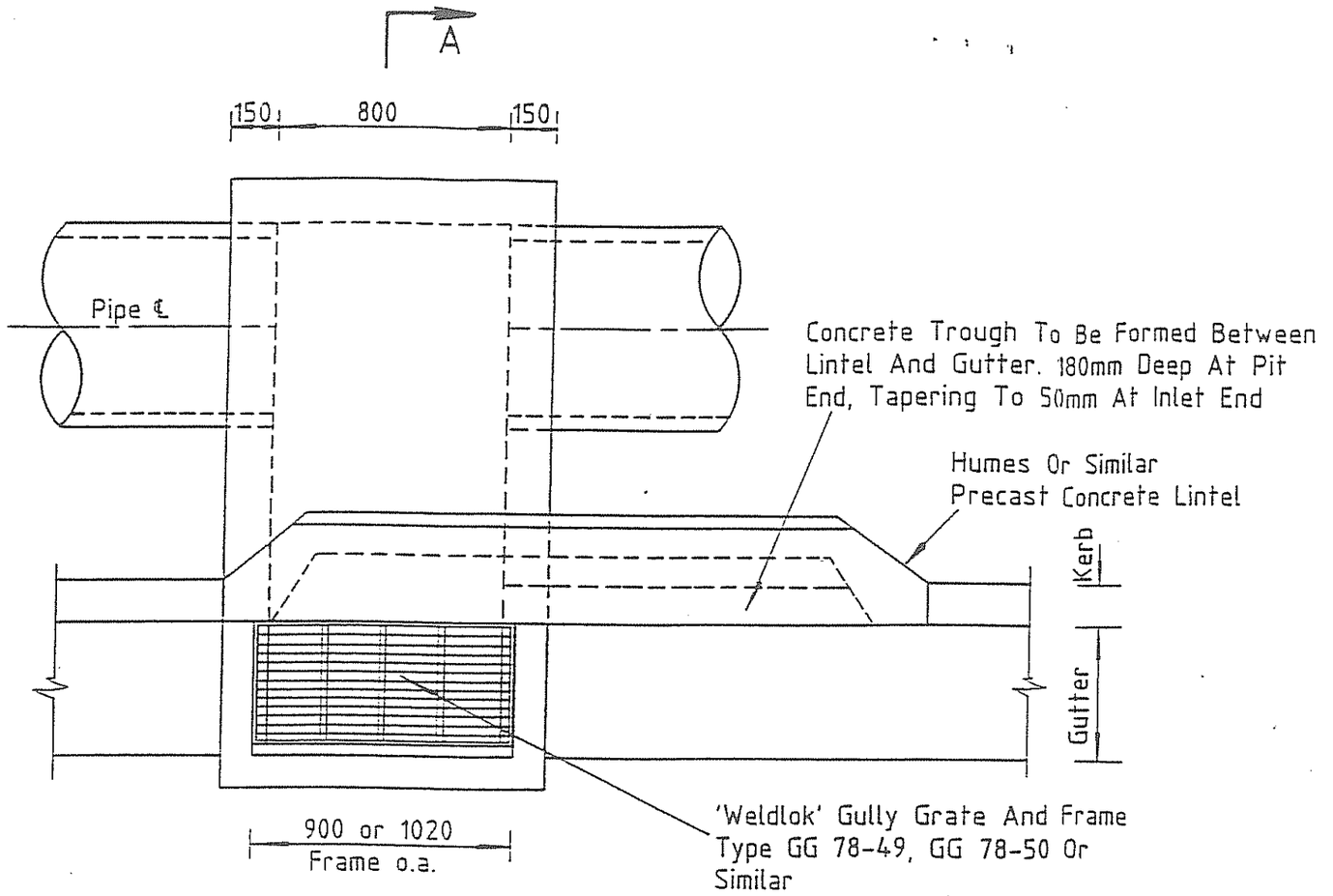
PIPE ℄ OFFSETS FROM KERB

PIPE ϕ	DISTANCE 'A'	PIPE ϕ	DISTANCE 'A'
375	100mm	750	200mm
450	100mm	825	250mm
525	100mm	900	300mm
600	150mm	1050	350mm
675	150mm	1200	450mm

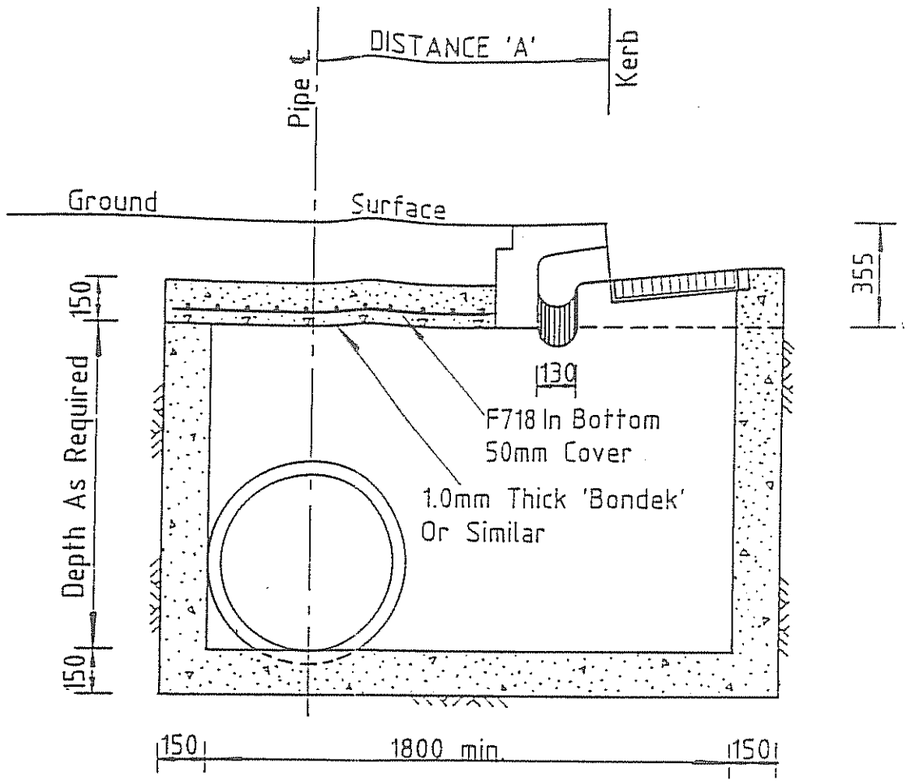
YOUNG SHIRE COUNCIL

KERB INLET PIT WITH GRATE
PIPE LOCATED UNDER K&G

SCALES	W & T SERVICES DIR.	DATE
AS SHOWN		
DRAWN		
TRACED		
CHECKED		
A3	DRAWING No. 6481	



PLAN



SECTION A-A

Note: For Pipes Greater Than 600mmϕ Extend Box Width In Increments Of 150mm To Suit Pipe Sizes

PIPE ℄ OFFSETS FROM KERB

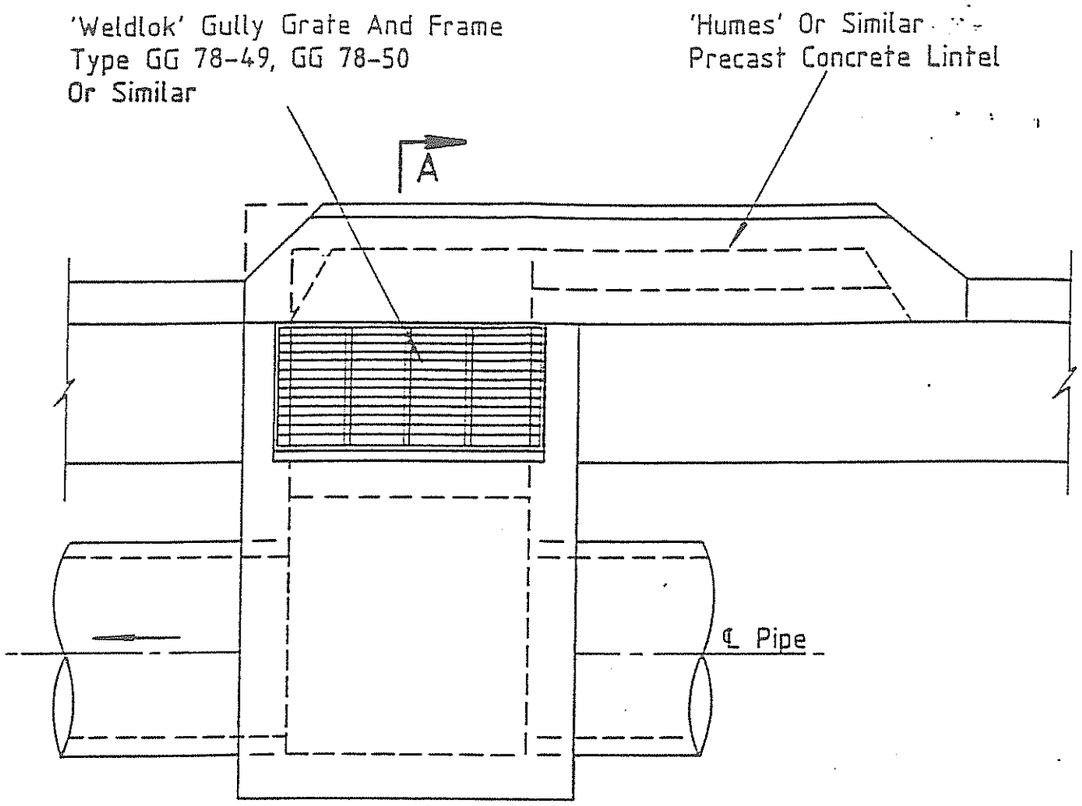
PIPE ϕ	DISTANCE 'A'	PIPE ϕ	DISTANCE 'A'
375	1000	750	1100
450	1000	825	1200
525	1000	900	1200
600	1000	1050	1300
675	1100	1200	1400

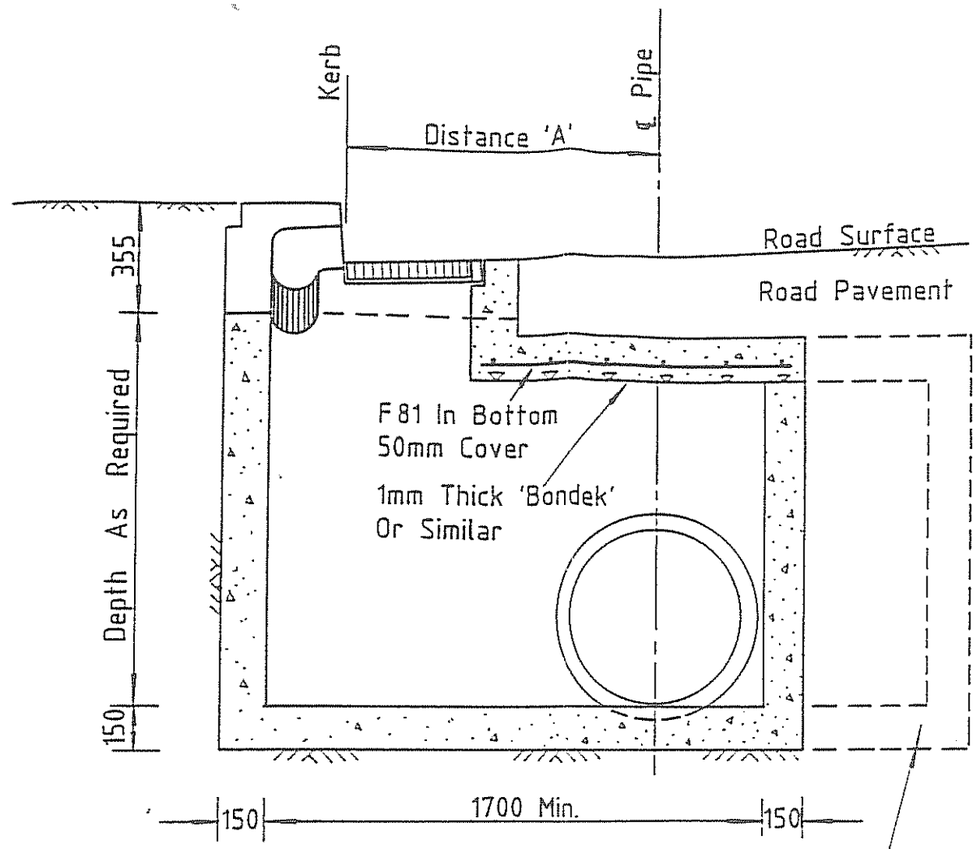
GENERAL NOTES

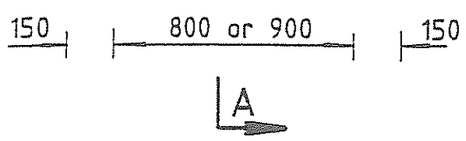
Concrete 20MPa

Step Irons To Be Provided @ 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

YOUNG SHIRE COUNCIL	SCALES AS SHOWN	W & T SERVICES DIR.	DATE
KERB INLET PIT WITH GRATE PIPE LOCATED BEHIND KERB	DRAWN		
	TRACED	A3	DRAWING No. 6482
	CHECKED		







Note: Kerb Inlet To Be Adjusted As Required To
Suit 'Weldlok' Grate & Lintel GG 78-57 Or Similar

PLAN

SECTION A-A

PIPE ℄ OFFSETS FROM KERB

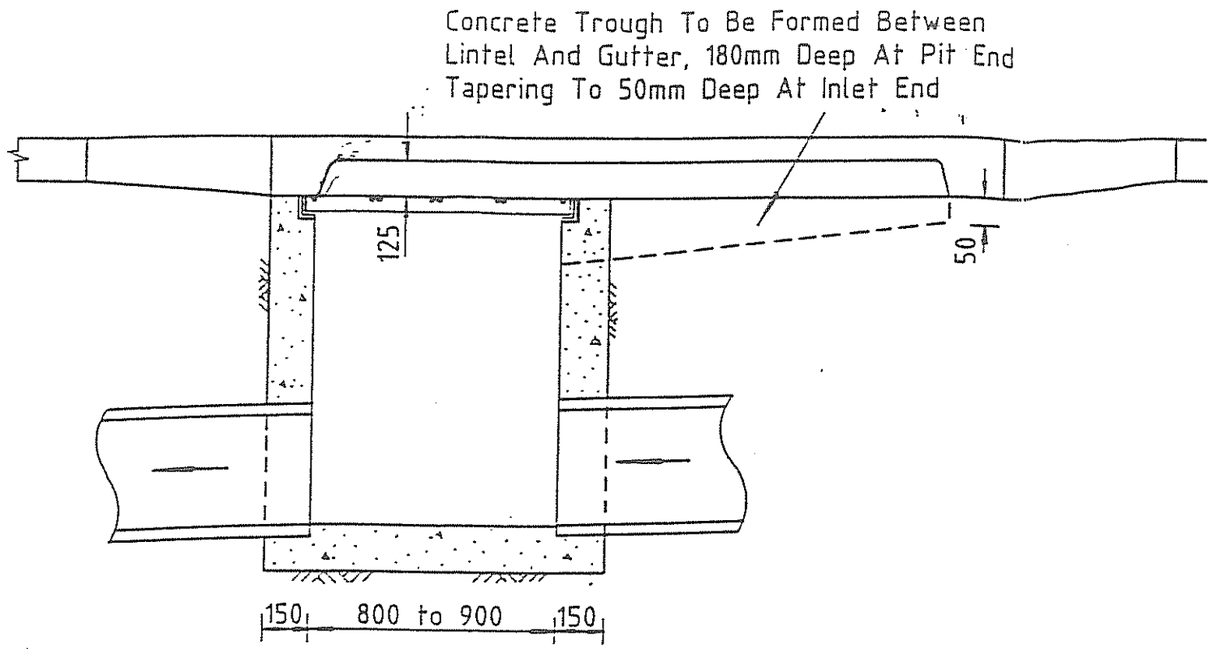
PIPE ϕ	DISTANCE 'A'	PIPE ϕ	DISTANCE 'A'
375	1100	750	1250
450	1100	825	1250
525	1100	900	1250
600	1100	1050	1450
675	1250	1200	1450

GENERAL NOTES

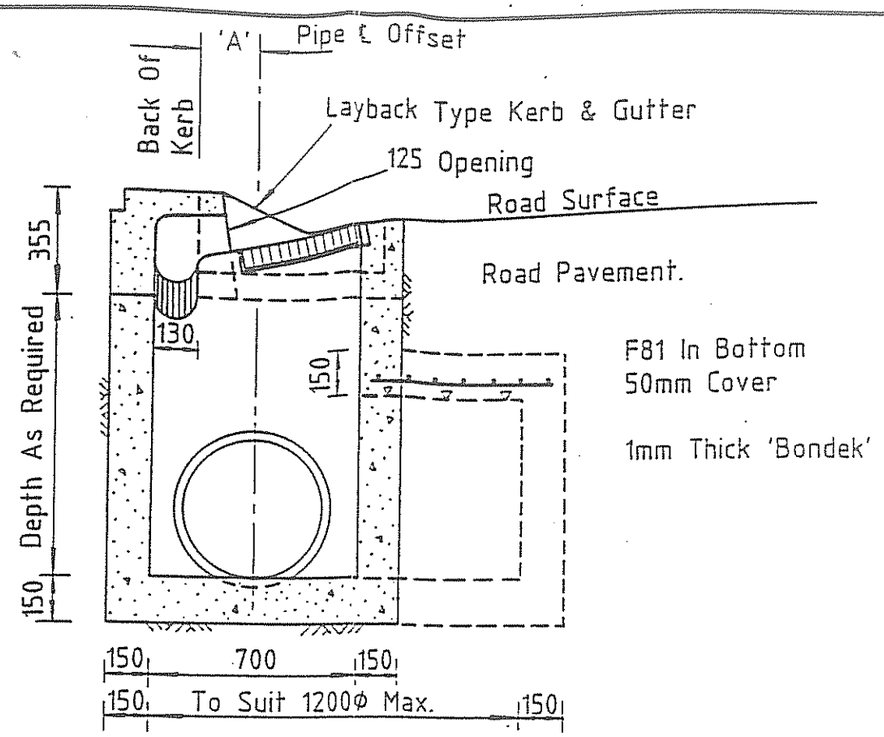
20MPa Concrete

Step Irons To Be Provided @ 300°/c
In Pits Deeper Than 1200mm

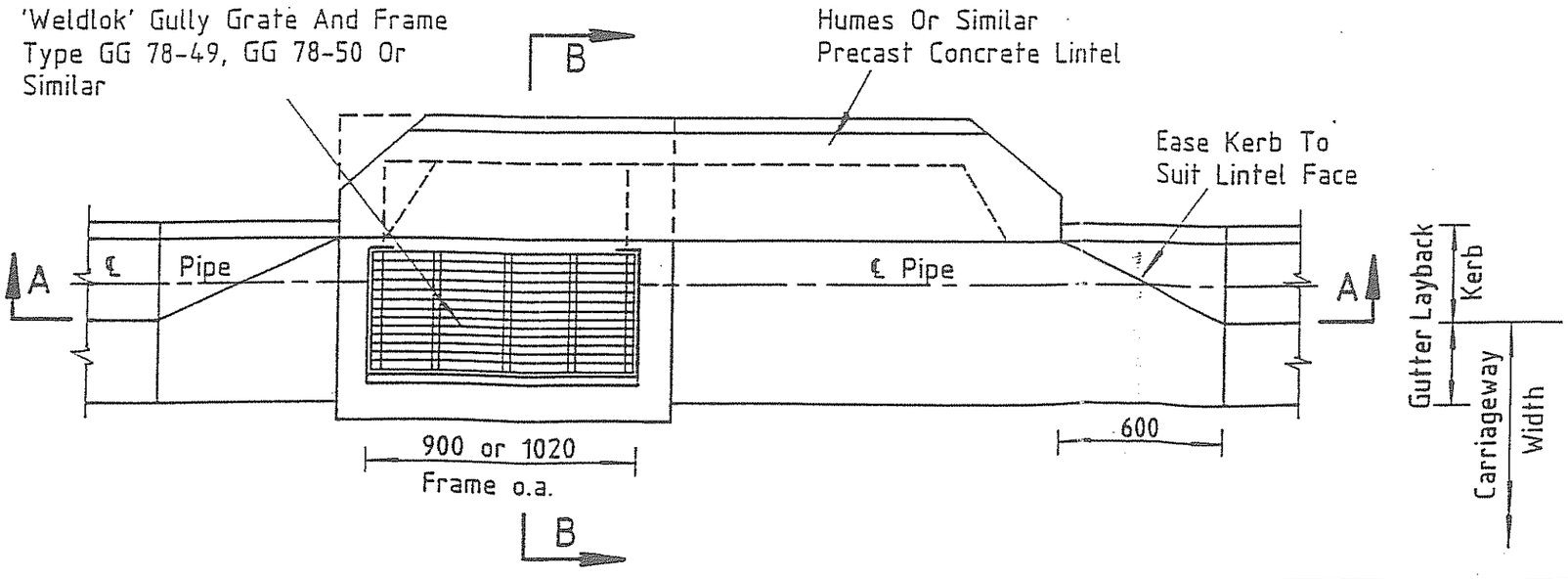
YOUNG SHIRE COUNCIL	SCALES AS SHOWN	W & T SERVICES DIR.	DATE
KERB INLET PIT WITH GRATE PIPE UNDER CARRIAGEWAY	DRAWN		
	TRACED	A3	DRAWING No. 6483
	CHECKED		



SECTION A-A



SECTION B-B



NOTE: Pit And Kerb Ease To Be Adapted To Suit 'Weldlok' Grate And Lintel GG 78-57 Or Similar

GENERAL NOTES

Concrete 20MPa

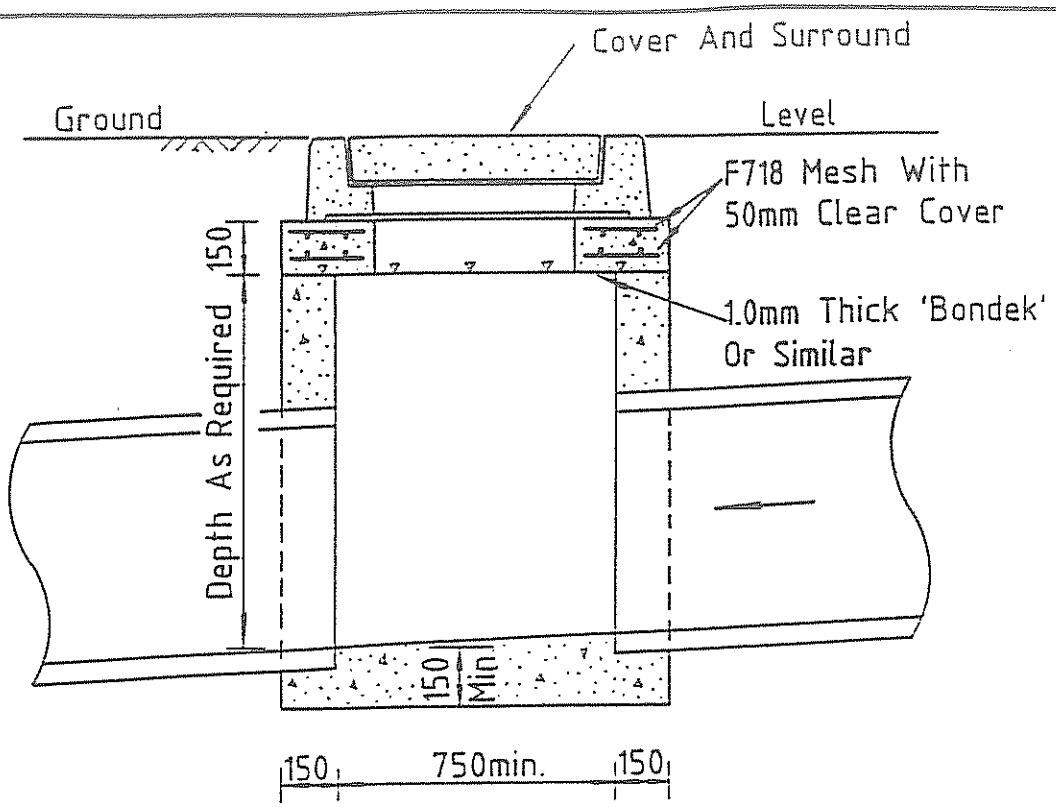
Step Irons To Be Provided @ 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

PIPE ϕ	DISTANCE 'A'	PIPE ϕ	DISTANCE 'A'
375	200mm	750	350mm
450	200mm	825	400mm
525	200mm	900	450mm
600	200mm	1050	500mm
675	300mm	1200	550mm

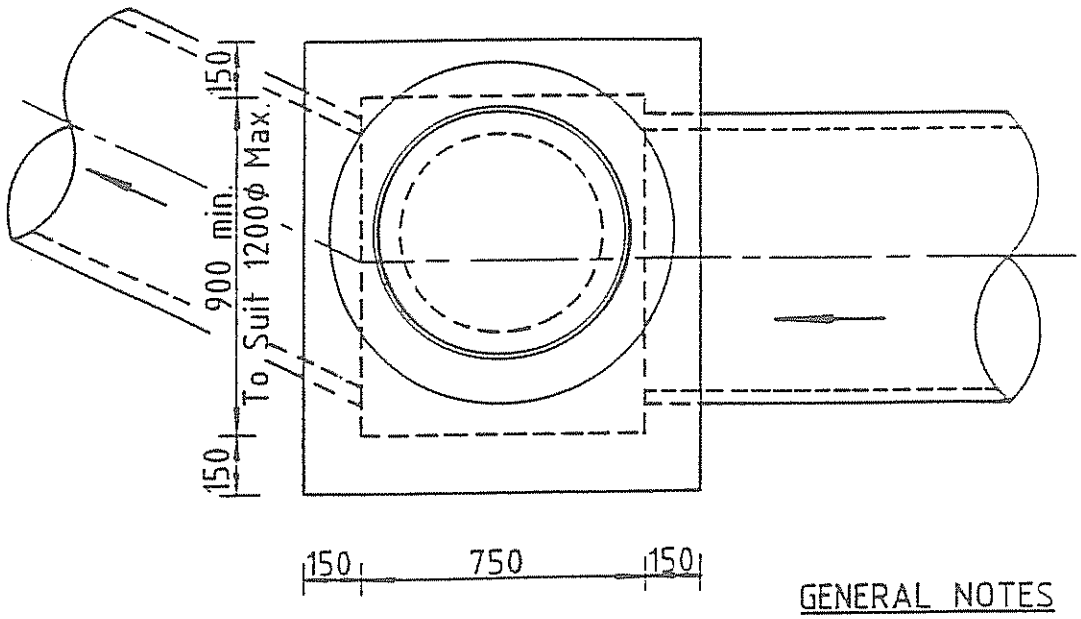
YOUNG SHIRE COUNCIL

KERB INLET PIT WITH GRATE FOR LAYBACK KERB & GUTTER

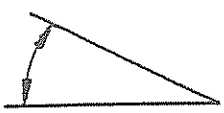
SCALES	W & T SERVICES DIR.	DATE
AS SHOWN		
DRAWN		
TRACED		
CHECKED		
A3	DRAWING No. 6484	



SECTIONAL ELEVATION



PLAN



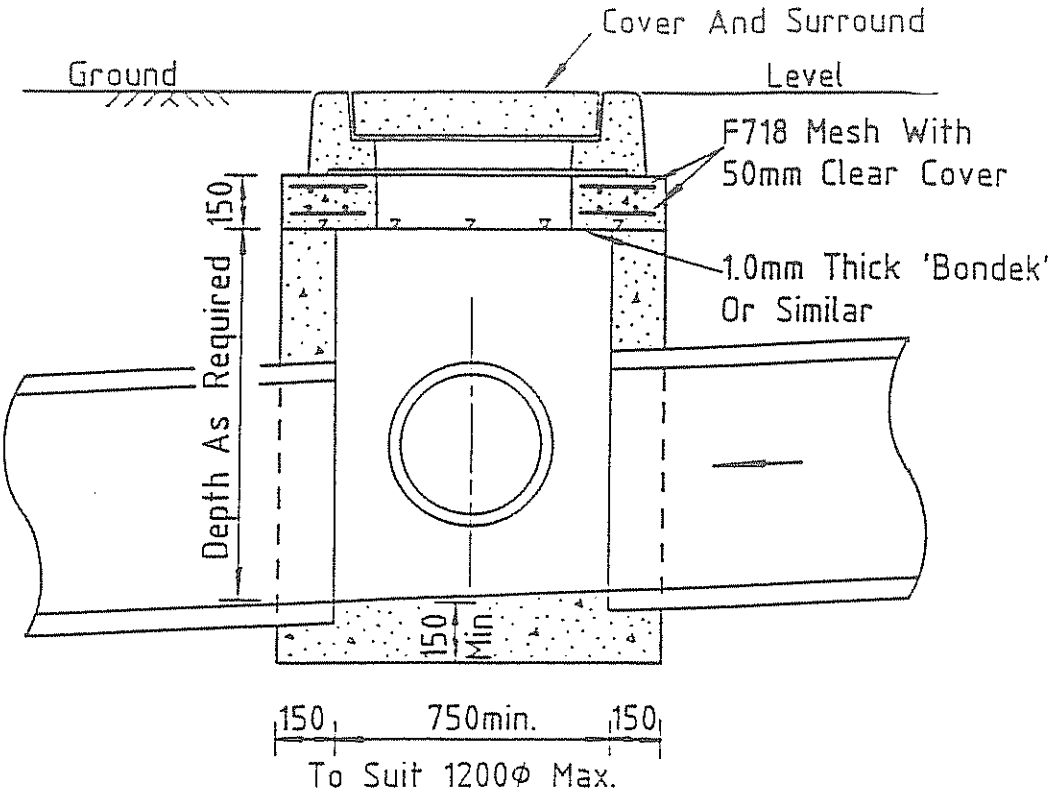
Angle As Required

GENERAL NOTES

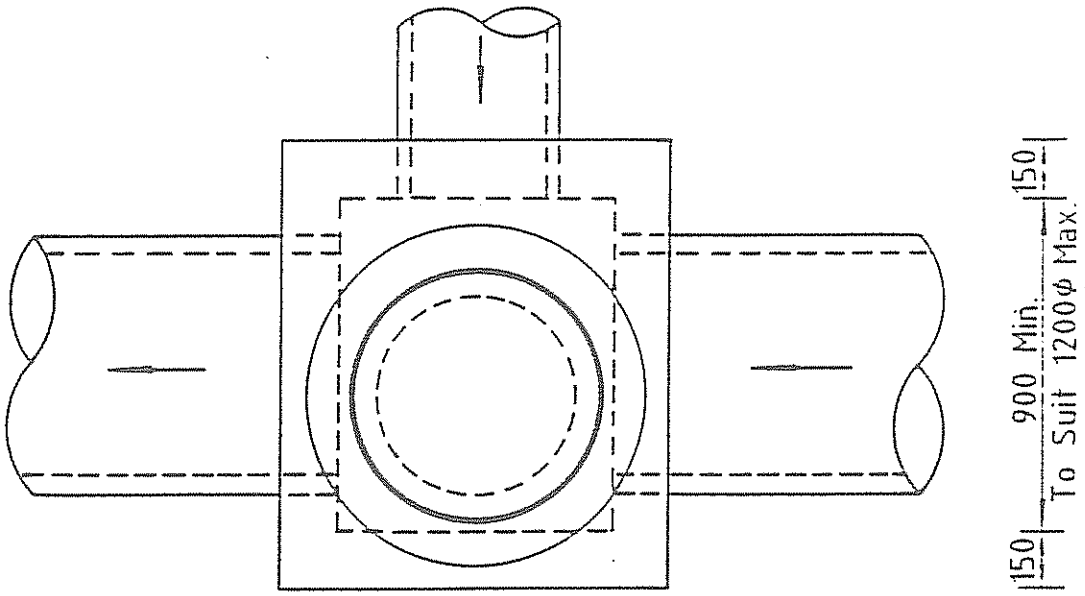
Step Irons To Be Provided @ 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

Concrete 20MPa

YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
ANGLE PIT	DRAWN	DRAWING No. 6485
	TRACED	
	CHECKED	



SECTIONAL ELEVATION



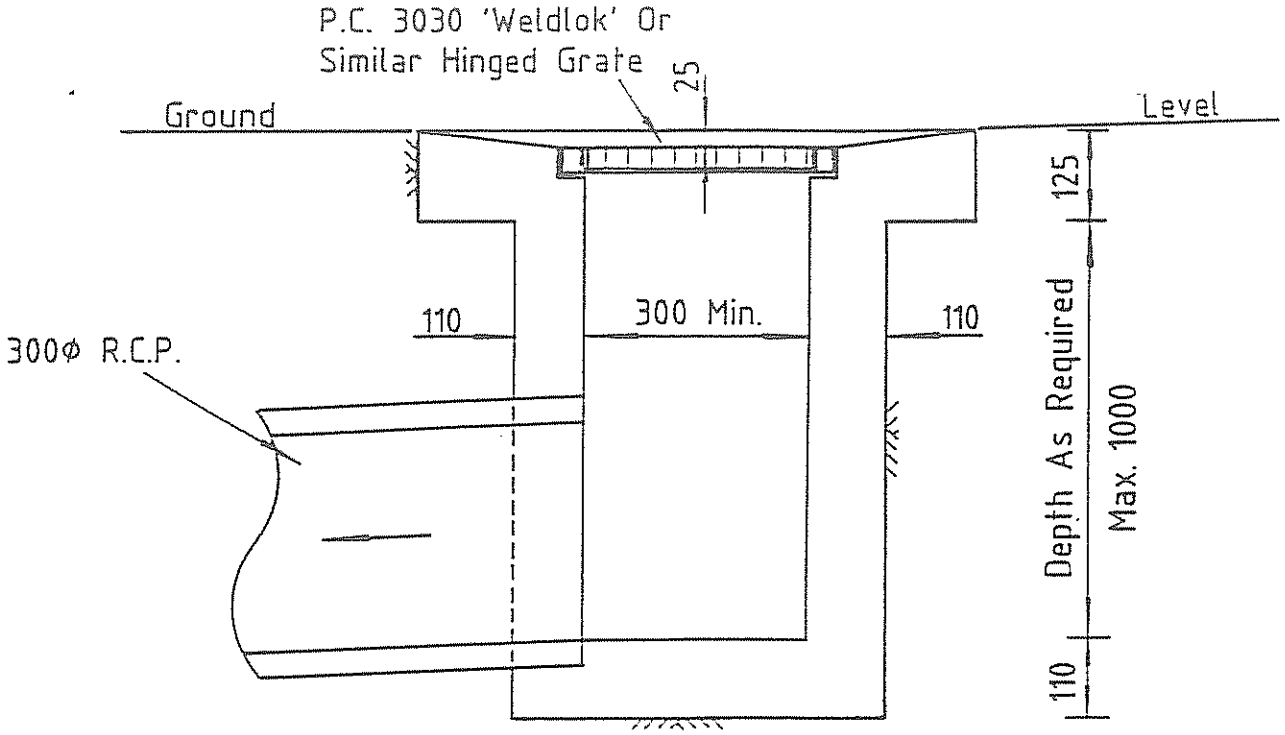
PLAN

GENERAL NOTES

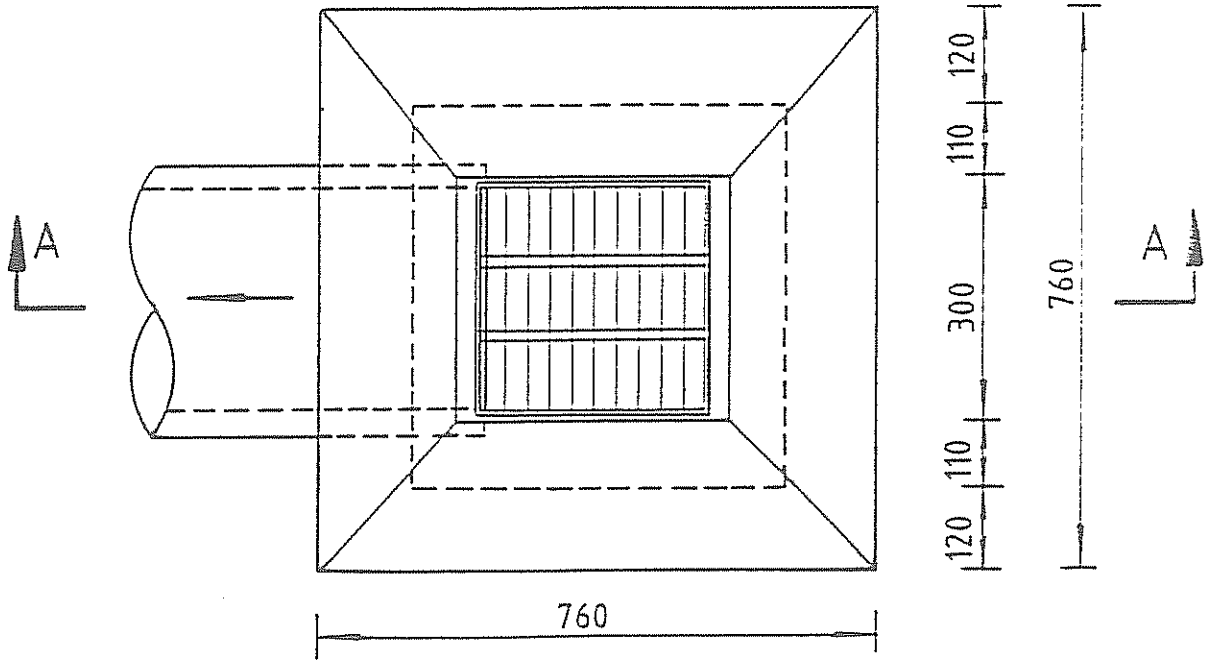
Concrete 20MPa

Step Irons To Be Provided @ 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
JUNCTION - DROP PIT	DRAWN	DRAWING No. 6486
	TRACED	
	CHECKED	



SECTION A-A



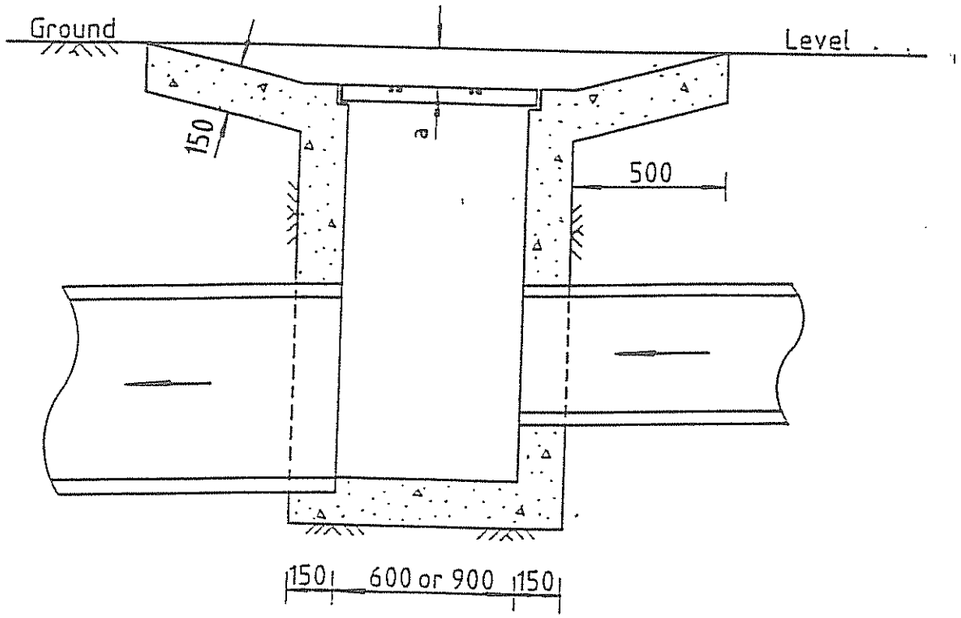
PLAN

GENERAL NOTES

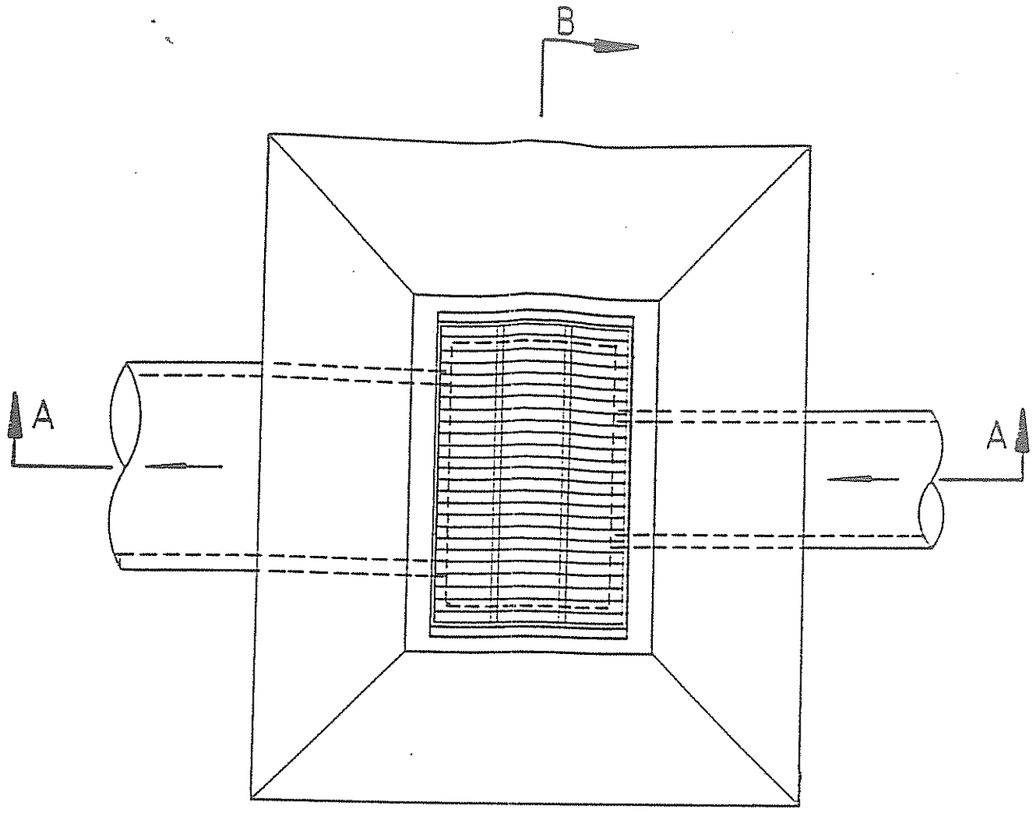
Concrete 20MPa

YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
FOOTPATH SURFACE INLET PIT WITH GRATE	DRAWN	DRAWING No. 6487
	TRACED	
	CHECKED	

a - Road Pavement 25mm
- Table Drain 100mm

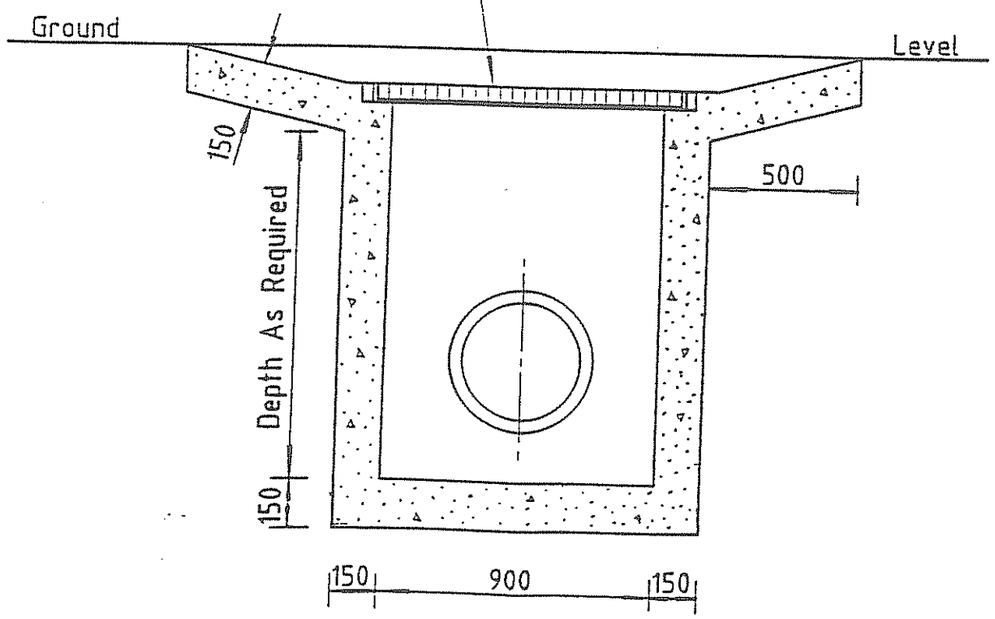


SECTION A-A



PLAN

'Weldlok' Surface Inlet Grate And Frame
Type P.C. 6090, P.C. 9090 Or Similar



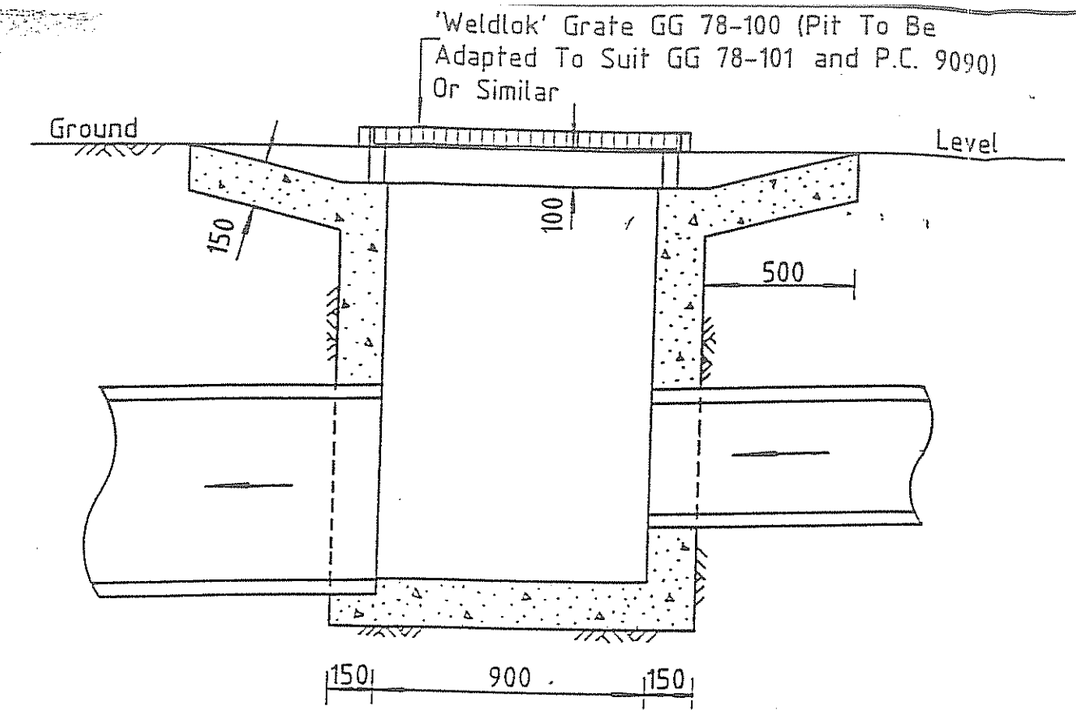
SECTION B-B

GENERAL NOTES

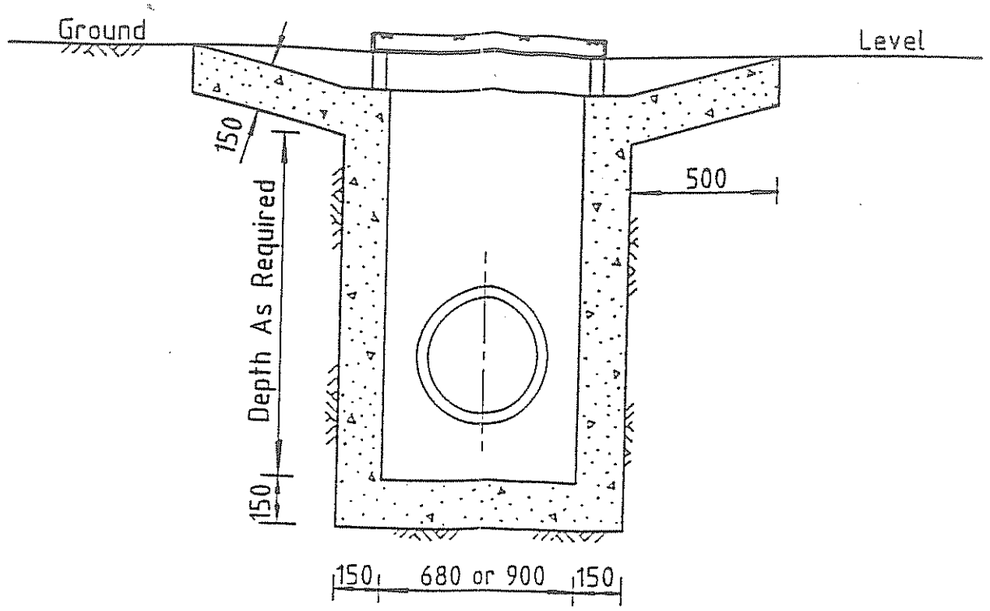
Concrete 20MPa

Step Irons To Be Provided @ 300$^{c}/_{c}$
In Pits Deeper Than 1200

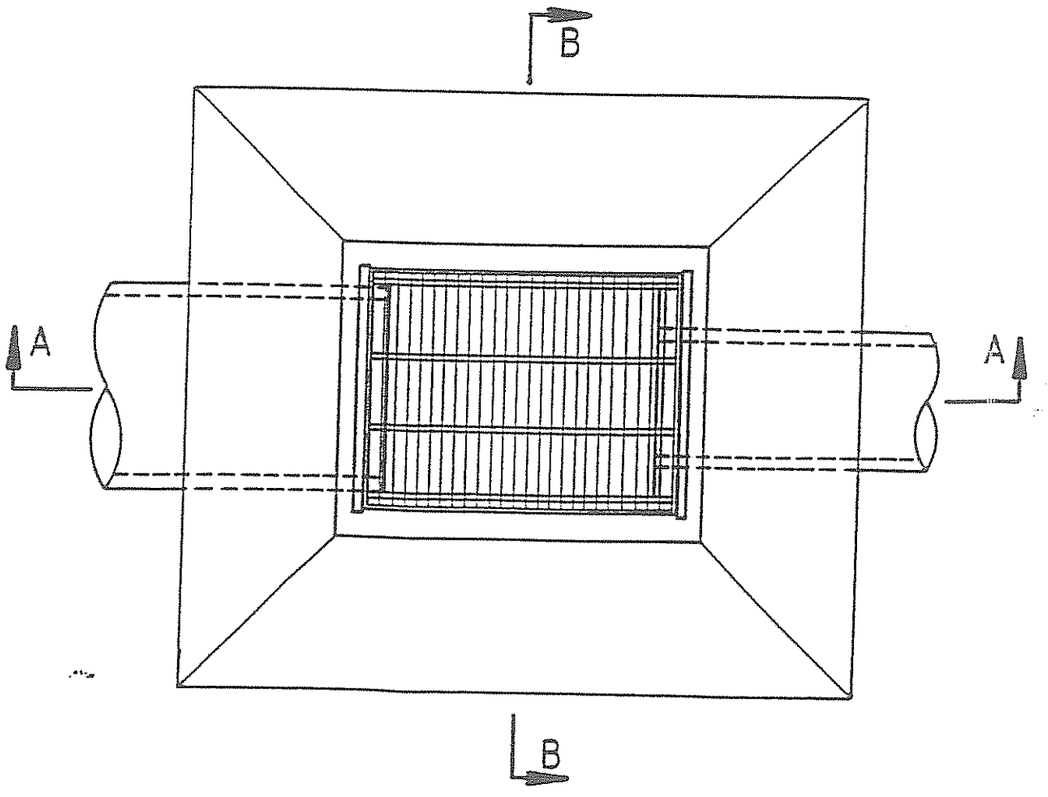
YOUNG SHIRE COUNCIL	SCALES AS SHOWN	W & T SERVICES DIR.	DATE
SURFACE INLET PIT	DRAWN		
	TRACED	A3	DRAWING No. 6488
	CHECKED		



SECTION A-A



SECTION B-B



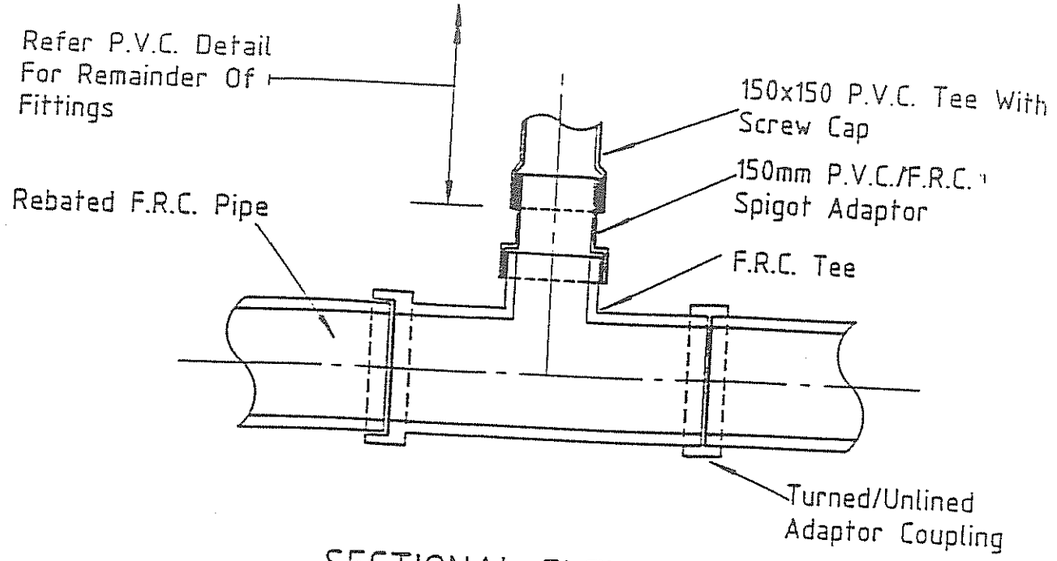
PLAN

GENERAL NOTES

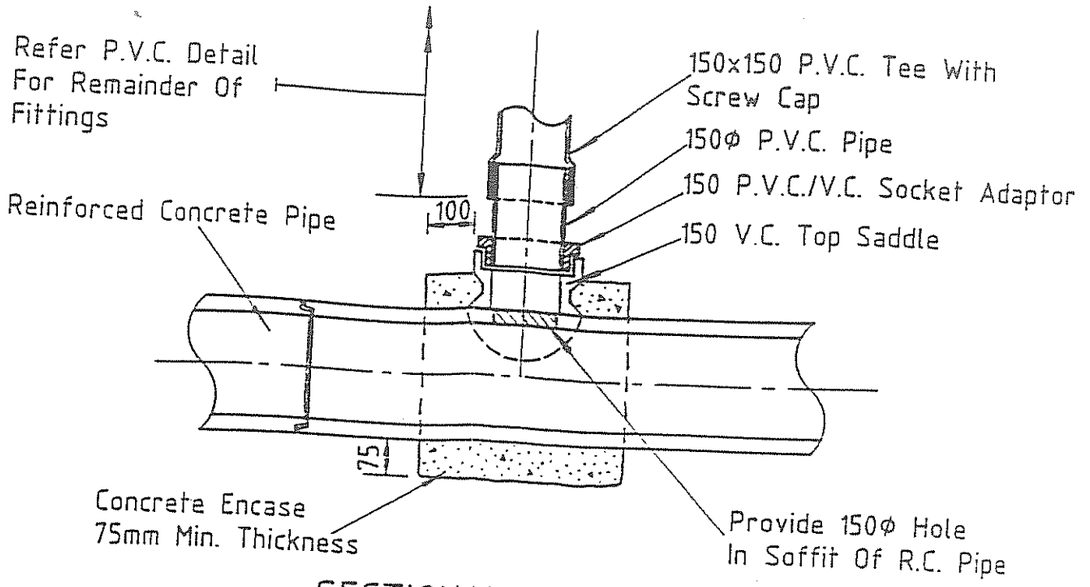
Concrete 20MPa

Step Irons To Be Provided At 300$^{c}/_{c}$ In Pits Deeper Than 1200mm

YOUNG SHIRE COUNCIL	SCALES AS SHOWN	W & T SERVICES DIR.	DATE
SURCHARGE SURFACE INLET PIT	DRAWN		
	TRACED	A3	DRAWING No. 6489
	CHECKED		

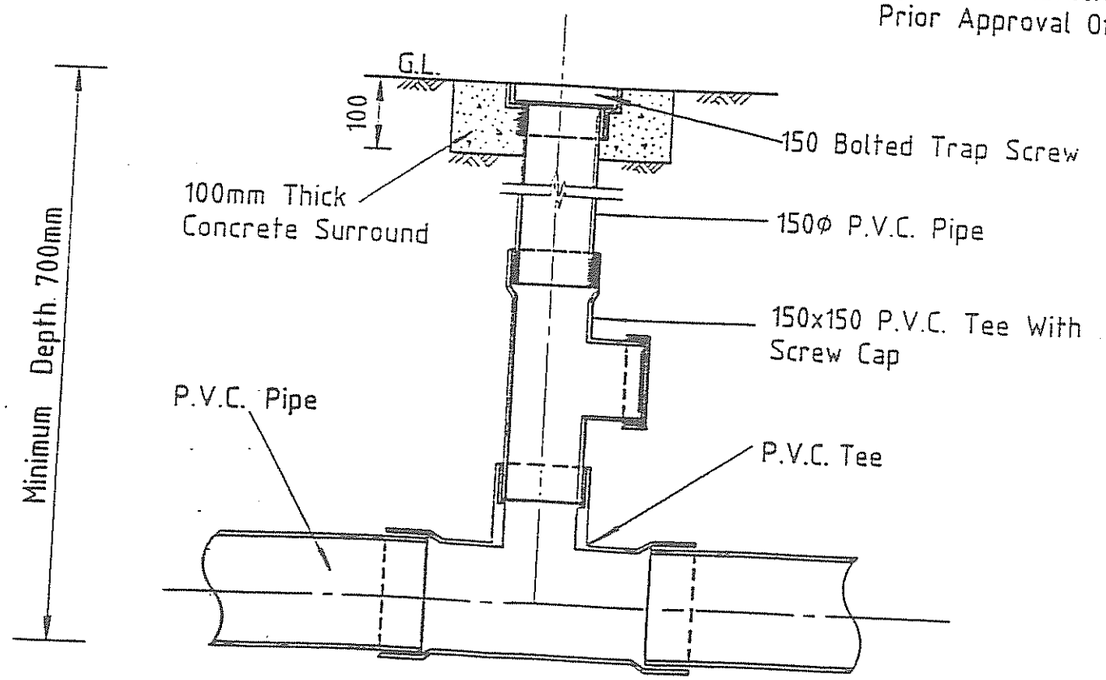


SECTIONAL ELEVATION
F.R.C. PIPE

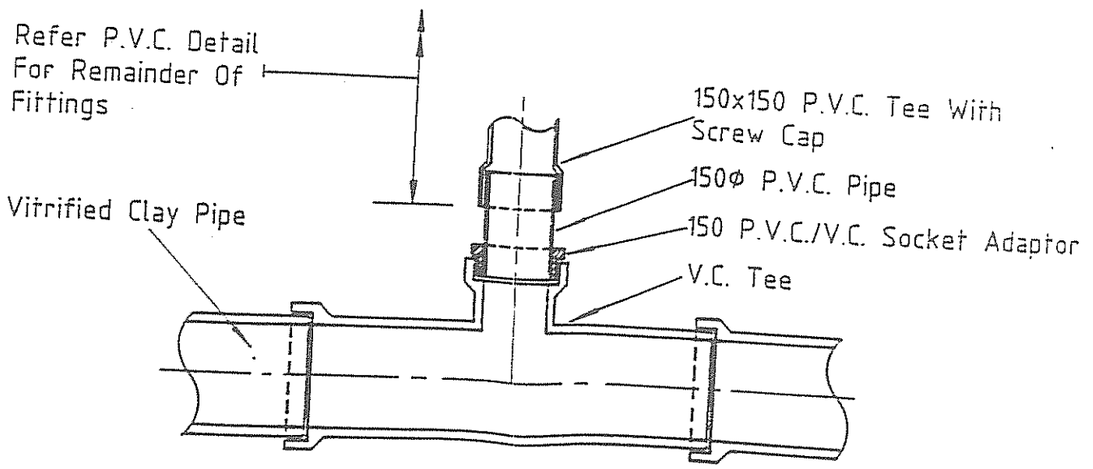


SECTIONAL ELEVATION
R.C. PIPE

Note:
Any Other Junction Connections Recommended By The Manufacturer Can Be Used With The Prior Approval Of The Director.

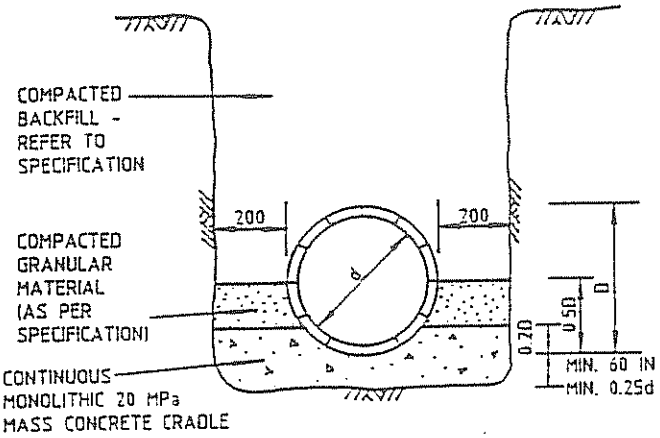


SECTIONAL ELEVATION
P.V.C. PIPE



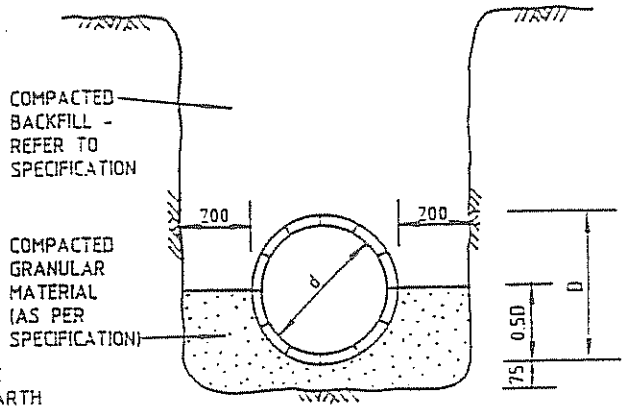
SECTIONAL ELEVATION
VITRIFIED CLAY PIPE

YOUNG SHIRE COUNCIL	SCALES AS SHOWN	W & T SERVICES DIR.	DATE
INTERLOT STORMWATER DRAINAGE INLET PITS	DRAWN		
	TRACED	A3	DRAWING No. 6490
	CHECKED		



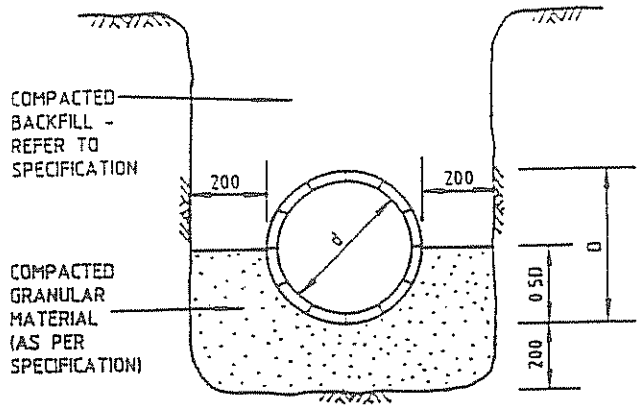
ROCK & EARTH FOUNDATION

PIPE BEDDING
TYPE A



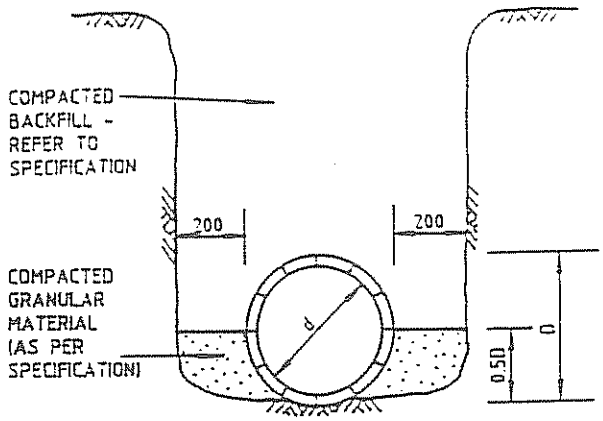
EARTH FOUNDATION

PIPE BEDDING
TYPE B (EARTH)



SHALE & EARTH FOUNDATION

PIPE BEDDING
TYPE D

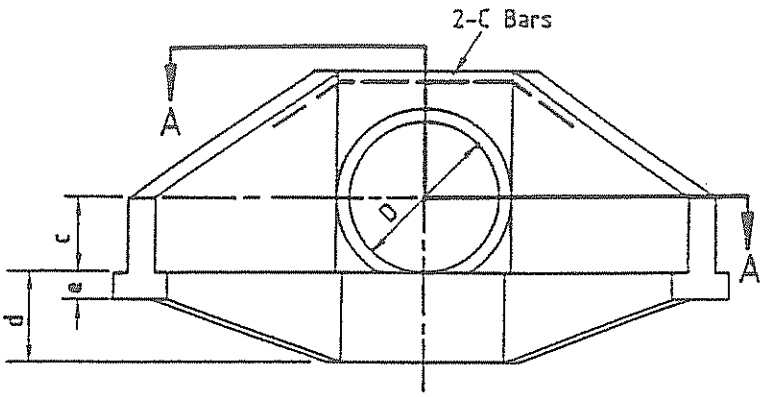


SAND FOUNDATION

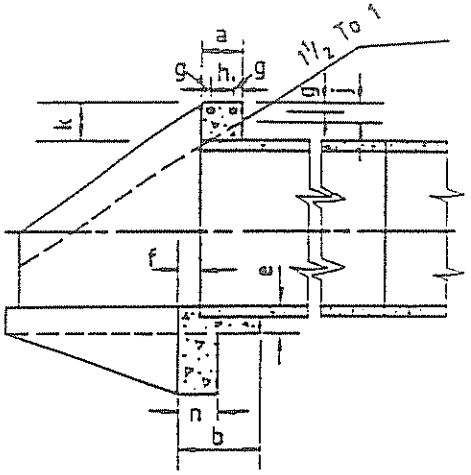
PIPE BEDDING
TYPE C

NOTE: THE ABOVE REQUIREMENTS COULD BE VARIED BASED ON THE MANUFACTURER'S RECOMMENDATIONS AND WITH THE PRIOR APPROVAL OF THE DIRECTOR.

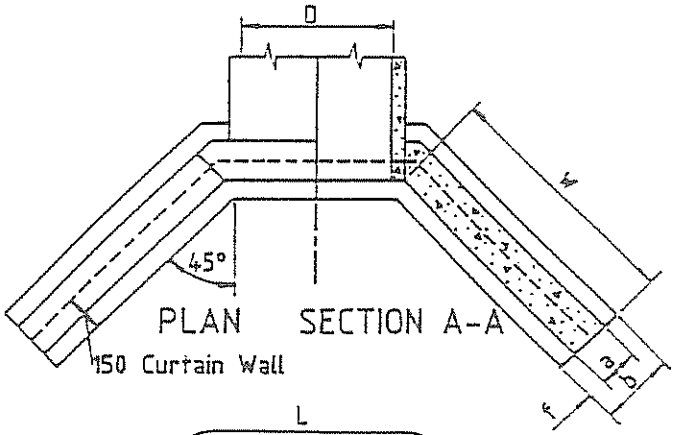
YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
PIPE BEDDING	DRAWN	DRAWING No.
	TRACED	7308
	CHECKED	

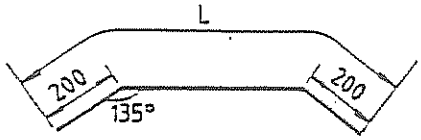


ELEVATION



SECTIONAL ELEVATION





C10 OR C12 BARS
2 Required Each Headwall

NOTES:

1. All Exposed Surfaces To Have 25 Chamfer
2. Reinforcing Bars To Be Cold Worked Deformed
3. Concrete Compressive Strength (F'c) 20 MPa At 28 Days
4. Precast Concrete Headwalls Of Sizes 375 To 750 (Incl.) May Be Substituted For The Equivalent Size In-situ Headwall

PIPE DIAMETER D	300	375	450	525	600	675	750	825	900
a	150	150	150	150	180	180	205	205	230
b	300	300	300	300	450	450	450	450	450
c	300	300	300	300	380	380	380	380	380
d	380	380	380	380	530	530	530	530	530
e	150	150	150	150	180	180	205	205	230
f	75	75	75	75	100	100	100	100	100
g	40	40	40	40	50	50	50	50	50
h	70	70	70	70	80	80	105	105	130
j	100	100	100	100	100	100	100	100	100
k	200	200	200	200	250	250	250	250	250
n	150	150	150	150	150	150	150	150	150
w	690	690	840	990	1120	1300	1450	1600	1780
L	800	840	915	990	1100	1200	1250	1350	1400
REINF. DIA.	10	10	10	10	12	12	12	12	12
REINF. LENGTH	1600	1680	1830	1980	2200	2400	2500	2700	2800
REINF., kg MASS	0.986	1.035	1.127	1.220	1.954	2.131	2.220	2.398	2.486
CONC. VOLUME m^3	0.25	0.27	0.33	0.38	0.67	0.85	1.02	1.22	1.40

ALL QUANTITIES ARE FOR ONE HEADWALL ONLY

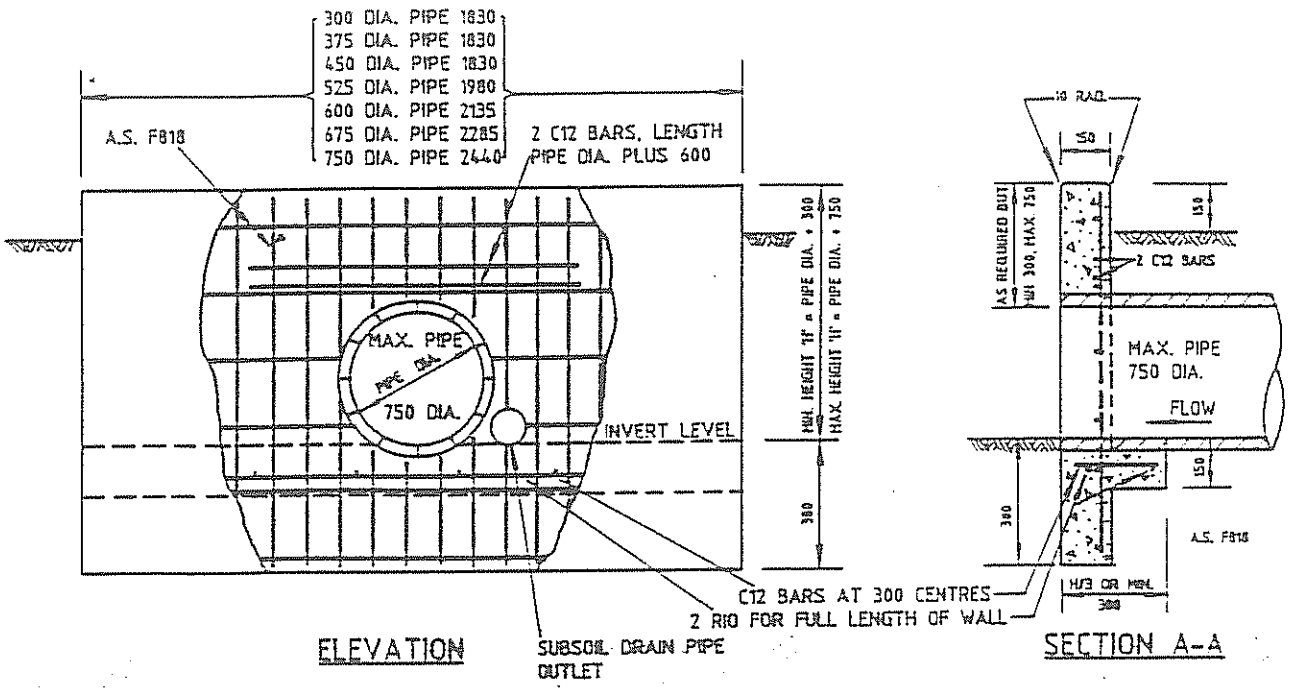
YOUNG SHIRE COUNCIL

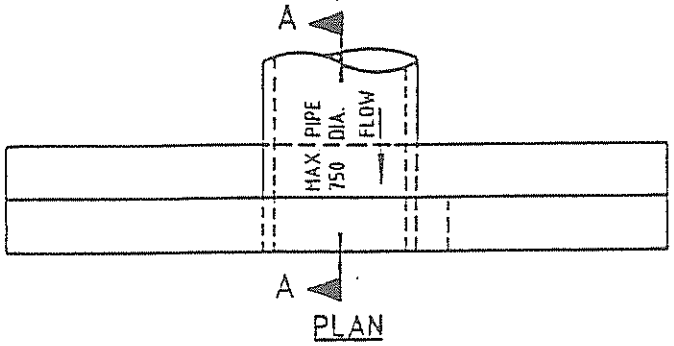
W & T SERVICES DIRECTOR | DATE

STANDARD CONCRETE HEADWALLS
FOR 300 TO 900 DIA. PIPES

DRAWN | TRACED | CHECKED

DRAWING No. 7310





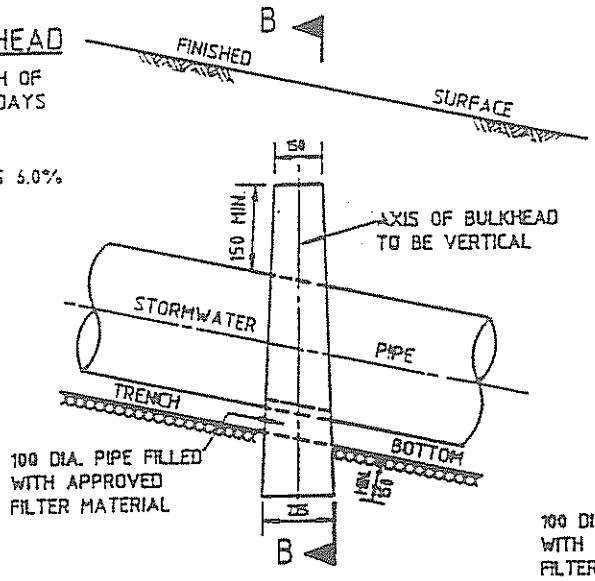
NOTES FOR CONCRETE HEADWALL

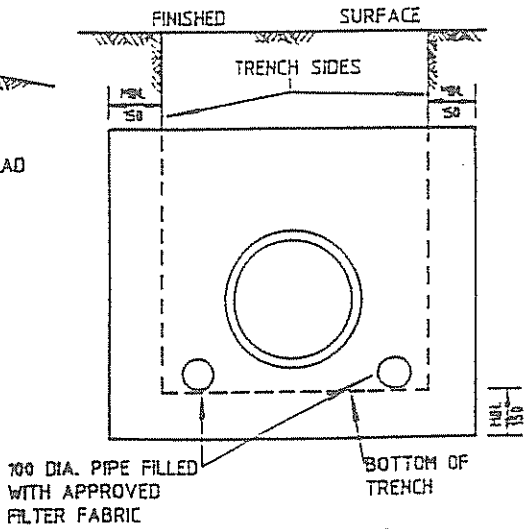
1. MINIMUM COVER OF CONCRETE OVER REINFORCEMENT TO BE 40.
2. COMPRESSIVE STRENGTH OF CONCRETE (F'c) AT 28 DAYS TO BE MINIMUM OF 20 MPa.
3. 100 DIA. SUBSOIL DRAIN PIPE 3000 LONG TO BE PROVIDED IN PIPE TRENCH AT PIPE INVERT LEVEL.

STRAIGHT CONCRETE HEADWALL FOR 300 - 750 DIA. PIPES

NOTES FOR BULKHEAD

1. COMPRESSIVE STRENGTH OF CONCRETE (F'c) AT 28 DAYS TO BE 20 MPa.
2. WHERE GRADE EXCEEDS 6.0%





CONCRETE BULKHEAD

YOUNG SHIRE COUNCIL	W & T SERVICES DIRECTOR	DATE
STRAIGHT CONCRETE HEADWALL CONCRETE BULKHEAD	DRAWN TRACED CHECKED	DRAWING No. 7312